

Study of runoff generation on hillslopes using tracer experiments and a physically-based numerical hillslope model

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Abstract Artificial rainfall and tracer experiments were conducted on two hillslope sites to identify runoff processes and to test a double porosity physically-based numerical hillslope model (QSOIL). The results show good agreement between the measured and the predicted runoff components, matric potential and water content. The spatial and temporal behaviour of the differ-system, resulting in bypass flow, and as the major contributing region for surface flow a highly permeable structured A horizon were attributed to one hillslope site. In contrast, the soil of the other site showed a bimodal pore-size distribution, assumed to be a combination of mesopores and micropores developed by plant roots, and resulting in a higher subsurface flow response.

INTRODUCTION

In the last 30 years, different approaches have been developed to describe runoff generation on hillslopes. Generally, runoff processes were observed by various field measurement techniques. Assuming a specific runoff controlling mechanism, rainfall-runoff models were developed for distinct time and space scales. However, a universally understanding of runoff processes was neglected, because rainfall-runoff models usually simplified the runoff generation. A process-orientated, physically-based model, however, is able to distinguish between different runoff processes, only if all flow processes are described and associated model parameters are measured.

The study of runoff generation occurring on 18 different hillslopes based on artificial extreme rainfall (Scherrer, 1997) provided insight into runoff processes. The mechanisms controlling runoff could not always be distinguished, although a physically based numerical hillslope model was used to predict discharge and soil moisture tension (Faeh, 1997). This model, QSOIL, can distinguish the following flow processes: surface flow, return flow, matrix flow, macropore flow and pipeflow. For this study, macropore flow is defined as water movement under small capillary forces vertically, while pipeflow is water movement laterally downslope. However, there are problems in estimating of model parameters. The objective of the present study, therefore, is to identify runoff generation processes on hillslopes by deriving the model parameters using all available information from field and tracer experiments, so that the usual parameter calibration is avoided.

METHODS

Field experiments

Two sites in the pre-Alpine Swiss research basin Vogelbach were chosen for the experimental work. The plots 60 m² each, were located on 17–25 steep hillslopes. The hillslope has gleysols which developed from Sardonna-Flysch and are influenced by the different vegetation covers, i.e. forest and grassland. A detailed description of the soils is given in Table 1. Two extreme rainfall events of 60 and 100 mm h⁻¹ were artificially simulated at each of the two hillslope research sites. In addition to the surface and subsurface runoff at the base of the plot and the subsurface pressure heads, water level and water content were measured and simultaneous tracer experiments were performed. Soil water changes were monitored during the experiments with soil moisture probes (TDR), tensiometers and piezometers. A side view of the forest site with the measurement devices is shown in Fig. 1. Tracer experiments with bromide (NaBr) at the grassland site and fluorescein (uranin) at both sites were conducted using different application and sampling techniques. As an example the flow, piezometer and tensiometer measurements taken during the second experiment at the forest site are shown (Fig. 2). The tensiometer response differed markedly and were affected by heterogeneities in hydraulic conductivities. No bypass flow, however, was detected. The breakthrough curves of fluorescein in the forest and grassland plots (Fig. 3) are similar to those expected for structured soils with different pore-size distributions.

Table 1 Properties of the experimental sites: *S* = sand content, *C* = clay content, *P* = porosity. Grassland has grass harvested for hay.

Properties	Location:	
	Grassland	Forest
Parent material	Sardona-Flysch	Sardona-Flysch
Soil classification	Gleysol on slopes	Gleysol on slopes
Texture: A horizon	<i>S</i> : 43%, <i>C</i> : 14%, <i>P</i> : 65%	<i>S</i> : 21%, <i>C</i> : 38%, <i>P</i> : 57%
B horizon	<i>S</i> : 27%, <i>C</i> : 28%, <i>P</i> : 52%	<i>S</i> : 15%, <i>C</i> : 40%, <i>P</i> : 53%
C horizon	<i>S</i> : 32%, <i>C</i> : 30%, <i>P</i> : 52%	<i>S</i> : 23%, <i>C</i> : 38%, <i>P</i> : 44%
Soil development	Development of highly porous, well structured A-horizon	Weak development of horizon
Humus	Mull	Mould
Content of coarse fragments	10–15%	15–25%
Size of coarse fragments	<40 cm (cherty and cobbly)	10–100 cm (stony)
Macropores	Earthworm holes, mouse-holes	Root channels
Slope	23°	17–25°

Physically-based numerical hillslope model (QSOIL)

A double porosity approach which links two domains, is used in QSOIL to describe flow in structured soils (Zuidema, 1985) because macropore flow generally has important effects on subsurface hydrology (Nielsen *et al.*, 1986). The first domain consists of the soil matrix, in which flow is described by the Richards equation. The

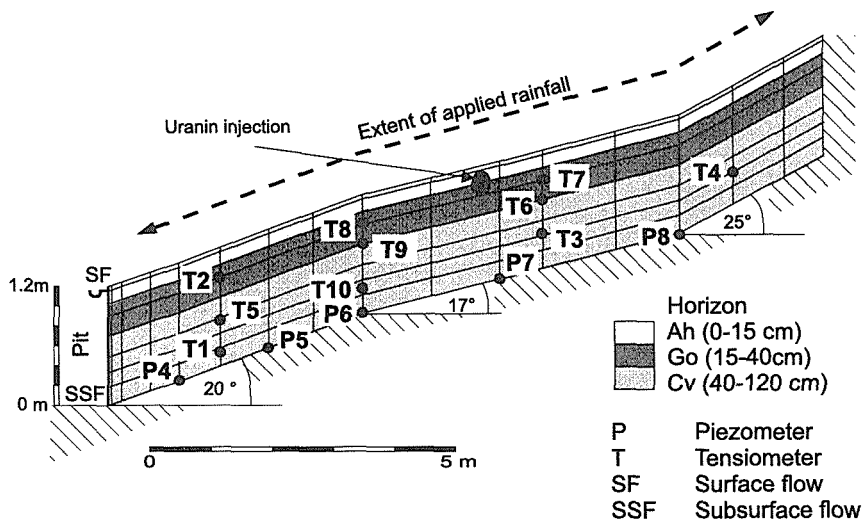


Fig. 1 Side view of the forest site with the measurement devices, point of uranin injection and discrete structure of the numerical hillslope model.

equations are solved for two dimensions by means of the Galerkin-type finite element technique. Dirichlet and Neumann boundary conditions are accounted for, enabling various seepage faces to be modelled. The second domain consists of macropores in which average flow properties (macro-porosity and flow velocity in macropores) are used to describe the system through a one-dimensional vertical diffusion wave process. The exchange of water between the two domains is computed by an interaction module. The water exchange depends on the hydraulic potential difference between the matrix and macropore domains. The hydraulic potential difference is derived from the macropore diameter and the hydraulic conductivity of the macropore lining and the surrounding matrix material. The global parameter describing the interaction and the macropore flow, however, cannot be estimated from general behaviour. The tracer experiments were used indirectly for the assessment. Overland and pipeflow are computed as one dimensional downslope kinematic or diffusion wave processes. The use of the two-dimensional QSOIL model is based on the assumption that little lateral water flows. The process elements that can be modelled are shown together with the parameters used by QSOIL in Fig. 4. For detailed information on QSOIL, see Faeh (1997).

Analysis of the tracer experiments

Different solute transport models were used in this study. In addition to the linear models, Convection-Dispersion-Model (CDM) and Transfer-Function-Model (TFM) (Jury, 1982), the non-equilibrium models, Single-Fissure-Dispersion-Model (SFDM) (Maloszewski, 1994) and Mobile-Immobile-Water-Model (MIM) (Van Genuchten & Wierenga, 1976), were fitted to the tracer breakthrough curves obtained from surface and subsurface runoff components, sampling tubes and single macropores in the pit. One model was selected for each breakthrough curve based on goodness-of-fit and

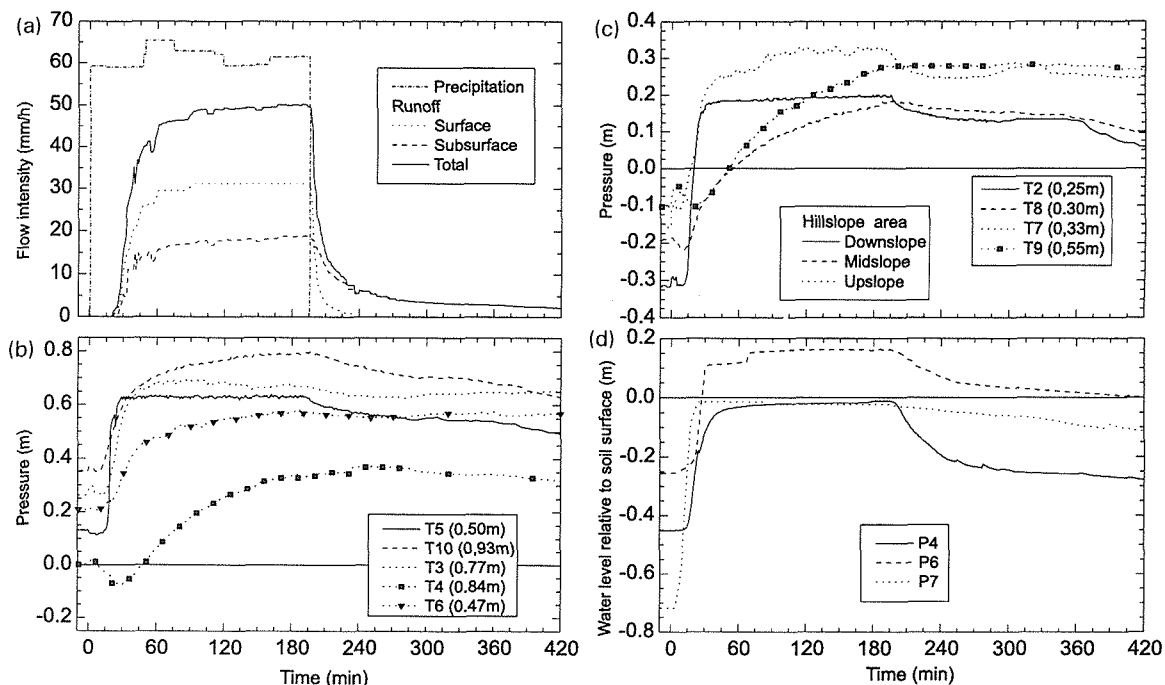


Fig. 2 Flow (a), tensiometer (b, c) and piezometer (d) measurements during the tracer experiment at the forest site (for the location of the measurement devices see Fig. 1).

the physical plausibility of the transport parameter. The agreement between predicted and measured tracer concentration of some breakthrough curves is shown in Fig. 5.

Parameter determination

The distributed model QSOIL requires information for homogeneous regions of the hillslope. To define homogeneous matrix flow elements, the soil properties are used as horizon-specific parameters. These are determined through pedo-transfer functions (Rawls & Brakensiek, 1989) that define the soil water retention curve and the relative hydraulic conductivity curve. To use QSOIL the previously derived parameters of the transport models and other field experiment results (water balance and bail tests) were synthesized for determination of indirectly measurable flow parameters required by QSOIL. The flow parameters include saturated conductivity for each horizon, which was determined from the effective porosity and pore water velocity and combined with the results of the bail tests and calculations using Darcy's law. The global macropore parameters were determined from the mobile pore water velocity, the effective porosity and the calculation of the macropore diameter using the Manning equation. This determination was a complex process because it considered all available information from the experiments and laboratory analyses. Nevertheless, the observed heterogeneity of the macropore flow at the hillslope scale makes it difficult to set one global parameter in QSOIL. The QSOIL parameters at both sites are listed in Table 2.

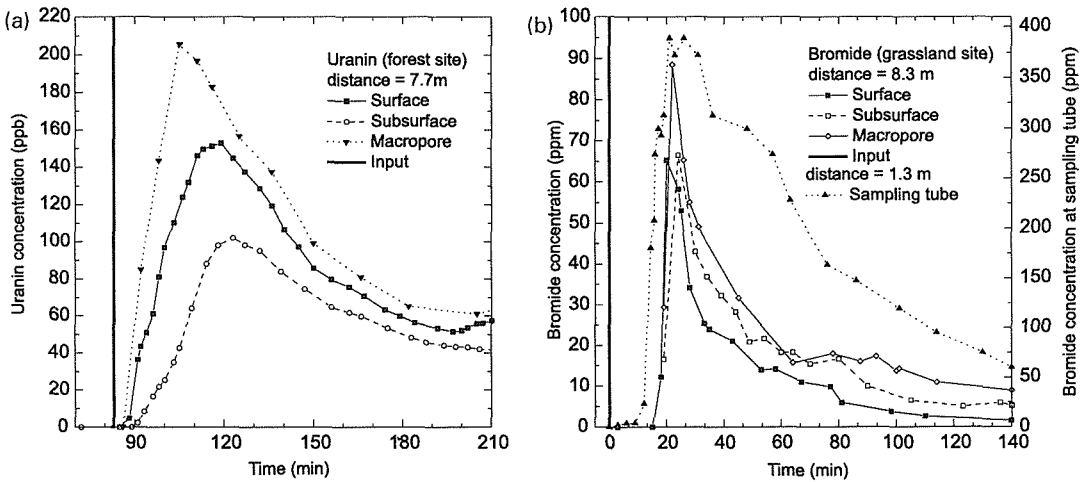


Fig. 3 Breakthrough curves of uranin during the second experiment at the forest site (a) and bromide during the second experiment at the grassland site (b).

RESULTS AND DISCUSSION

The computations with the QSOIL parameter values were derived from field and laboratory data and not from calibration with both the discharge and pressure change measurements. The simulations of each site reveal important differences in their runoff processes.

Grassland site

The transport parameters (Table 2) for the grassland site show realistic values for the macropore domain and are consistent with field observations of earthworm holes, the major origin of vertical macropores, and mouse-holes, the major origin of horizontal macropores. The simulation results describe the measured matrix potential well. The

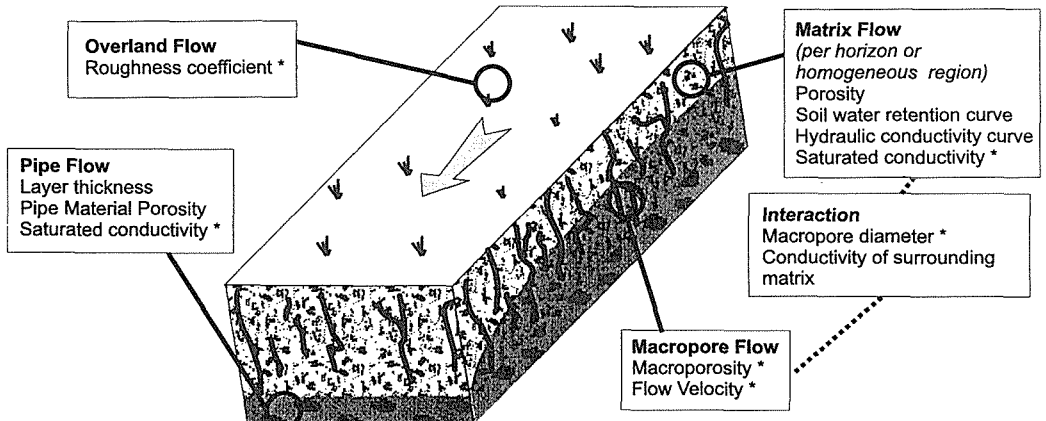


Fig. 4 Flow process elements of the QSOIL model with their respective modelling parameters. The parameters marked with * are indirectly determined using the information of the tracer experiments.

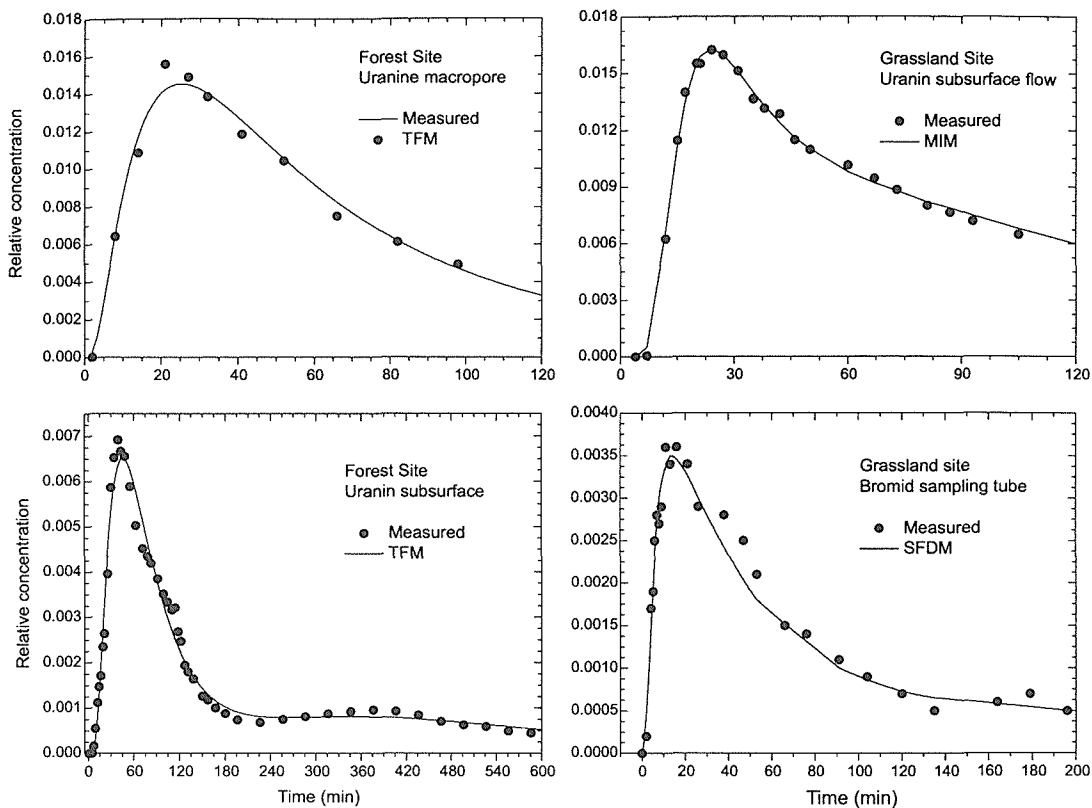


Fig. 5 Examples of measured and fitted breakthrough curves for the forest site, using TFM, and the grassland site, using MIM and SFDM.

calculated surface and subsurface flow components match the measured data especially well. Field observations already suggested the importance of the upper soil horizon (A horizon) as the major contributing region to the measured surface flow. Actually, 70% of the simulated surface and near-surface flow originates from this horizon. Runoff generation was mainly controlled by the highly permeable A horizon and the strongly developed vertical macropore system in deeper horizons, which results in bypass flow.

Table 2 QSOIL flow parameters determined by analysing the results of rainfall and tracer experiments and soil properties.

Location	Saturated conductivities (m s^{-1})			Global macropore parameters			Overland flow
	vertical per horizon	horizontal per horizon		Flow velocity (m s^{-1})	Effective porosity (%)	Diameter (mm)	
Grassland	2×10^{-5}	3×10^{-6}	3×10^{-6}	0.006	0.6	0.5	4
	1.7×10^{-3}	5×10^{-5}	2×10^{-5}	0.04*	10.0*	3.0*	
Forest	1×10^{-5}	1×10^{-5}	4.4×10^{-7}	0.0015	4	0.07	1
	8×10^{-5}	5×10^{-5}	4.4×10^{-7}				

* Horizontal values in A-horizon at grassland site.

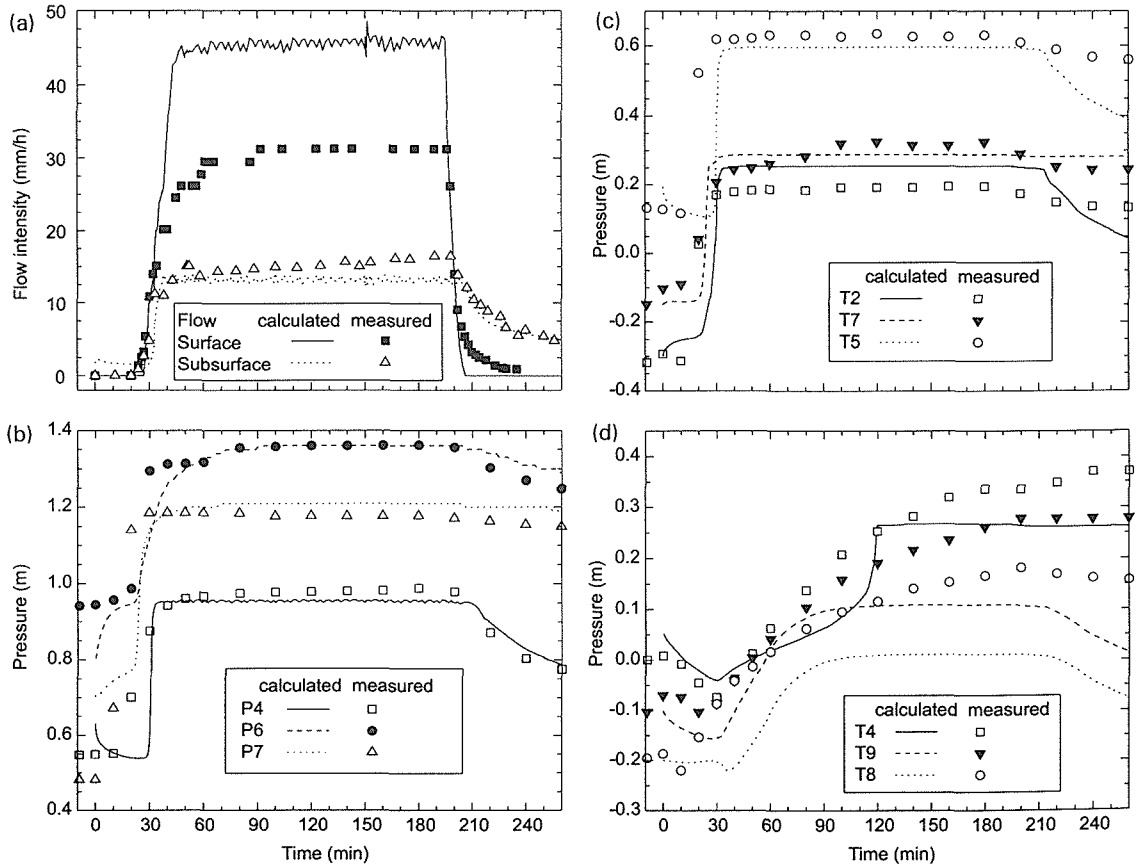


Fig. 6 Simulation results for the forest site using a bimodal pore-size distribution (for the location of the measurement devices see Fig. 1).

Forest site

The infiltration process and the development of the macropore structure of the forest site differs from the grassland site. The high effective porosity (4%) and the low effective pore diameter (70 μm) of forest site macropore system differs significantly from those of the grassland site (Table 2). The pore system is developed by plant roots, which form a dense lateral and vertical net of pore-channels, having mainly small diameters. The analysis of the breakthrough curves supports this hypothesis because of the high diffusion of the tracer from the mobile to the immobile phase. The high diffusion implies a pore domain with a small pore diameter and thus strong capillary forces and no hydraulic separation between the two pore domains. Additionally, there was no bypass flow observed at this site. Consequently, the soil at the forest site can be described by a bimodal pore-size distribution, assumed to be a combination of mesopores and micropores (Wilson *et al.*, 1990). The results of the QSOIL simulations using these assumptions are shown in Fig. 6. The different responses of the tensiometers and piezometers are well simulated, but the surface flow, however, is overestimated, probably due to lateral losses after the soil is

Table 3 Observed relation of the experimental site and flow paths to the suitable transport and flow models for different pore-size distributions.

Experimental site <i>Flow path</i>	Pore-size distribution	Transport model	QSOIL
Grassland <i>Subsurface flow in Go/C horizon</i>	Unimodal	CDM	Matrix flow
Forest <i>Infiltration and subsurface flow</i>	Bimodal	TFM	Matrix flow with properties for a bimodal pore-size distribution
Grassland <i>Infiltration and subsurface flow in A-horizon</i>	Matrix and well developed macropore system	SFDM MIM	Matrix flow macropore flow (pipeflow)

saturated. Consequently, a relation for the selection of the suitable transport and flow models for different flow paths could be developed (Table 3).

CONCLUSION

On the condition that measurements and simulations of runoff processes are conducted in the same spatial and temporal resolution, the combination of artificial rainfall and tracer experiments provides a suitable tool for identifying and characterizing discharge formation processes. This method also supplies information to determine difficult to measure parameters for QSOIL. There are, however, limits to the extent to which distinct microscale components and processes can be described. The hillslope-scale does not seem suitable for the identification of a single microscale process, because the simulations are based on so-called “effective parameters” derived from tracer experiments. The transformation of transport parameters into flow parameters is critical if the structure of the fitted solute transport model does not account for real physical flow mechanisms. To minimize these difficulties, future research should focus on developing transport and flow models that are linked to each other. Furthermore, there is a need for the physical flow descriptions of the matrix and macropore flow to be validated at different scales.

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7 Terrestrial Ecology and Limnology

Synergetic effects of organic pollution and river slope variability on the biotic continuum of the Adige River (south Tyrol, Italy)

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Abstract The Adige River (south Tyrol, Italy) can geomorphologically be divided into two main sections because of a knickpoint located 70 km downstream from the source. Immediately after the knickpoint the stream receives an untreated urban sewage effluent. Along the two sections different in hydrological, morphological and food resource features, five sites were investigated. Macroinvertebrate samples and environmental parameters were seasonally analysed for two years (1995–1996). Taxonomical as well as functional differences in macroinvertebrate community composition were found between the upper and lower river section. The biotic discontinuity was mainly shown by the functional distribution patterns, reflecting the differences in substrate heterogeneity and food resource features in the two sections.

INTRODUCTION

In stream ecosystems, environmental gradients often follow longitudinal patterns. As lotic communities are adapted to utilize the structural and resource features of a specific habitat, they provide a basis upon which hypotheses, relating abiotic factors to community structure and function, can be tested. Some of the main parameters affecting the habitat are geomorphological features such as stream-bed slope, substrate composition and current velocity (Cummins & Lauff, 1969; Clup & Davies, 1982; Cooper & Knight, 1985; Minshall *et al.*, 1985; Richards & Bacon, 1994). Sharp slope variations mostly occur in the upper part of the river, characterized by canopy shading and substrate heterogeneity, while the lower part has weak slope variations, almost no canopy shading and fine sediment structures (Vannote *et al.*, 1980; Minshall *et al.*, 1985). Consequently, energetic conditions also change longitudinally along undisturbed stream gradients. Allochthonous detritus is the major energetic resource for streams with prevalent heterotrophic character (Cummins, 1992). The unidirectional flow is responsible for downstream transport of the organic matter through longitudinally linked subsystems, by the transition from coarse particulate organic matter (CPOM) to fine particulate organic matter (FPOM). The process of retention, which includes the deposition and trapping of organic matter, provides a critical link among input, long-term storage and processing, especially of CPOM. The retentive capacity of streams is a function of hydrology, substrate composition and stream-bed morphology.

Increased loading of nutrients in streams can threaten the ecological integrity of the ecosystems. This could happen for example as a consequence of sewage runoff.

Moreover the capacity of a stream to transport the input surplus is often reduced because of other human impacts on stream morphology, such as rectification or manipulation of riparian areas.

Lotic macroinvertebrates reflect the synergetic effects of all variables influencing their habitat. Thus the community structure and function can be seen as biotic response allowing a global overview (Cummins, 1992).

Due to a knickpoint the upper part of the River Adige, studied in this work, can geomorphologically be divided into two main sections. Immediately after the knickpoint the stream receives untreated urban sewage.

The aim of this study is to evaluate the synergetic effects of different geomorphological conditions and the input of the sewage effluent on the river continuum, by analysing the response of the macroinvertebrate community structure taxonomically as well as functionally.

MATERIALS AND METHODS

Study area

This research was conducted in the upper part of the Adige River (south Tyrol, Italy). The drainage basin of this lotic environment is 11 954 km², the third largest in Italy. The Adige, the second longest Italian river (409 km), has its source at 1586 m a.s.l. and flows into the northern Adriatic Sea.

Five sampling stations were chosen for the examination of the macroinvertebrate community along the river continuum. The first three stations are separated from the other two by a knickpoint located 70 km down from the source. The upper river section is characterized by slope variations and heterogeneous channel morphology, while the lower section is characterized by almost absent slope variations and homogeneous channel morphology, together with rectification of the river margins for almost the entire analysed zone. Furthermore, after the knickpoint the river receives an untreated urban sewage effluent from the city of Merano (Fig. 1). The riparian vegetation mainly consists of *Populus* spp. and *Salix* spp.

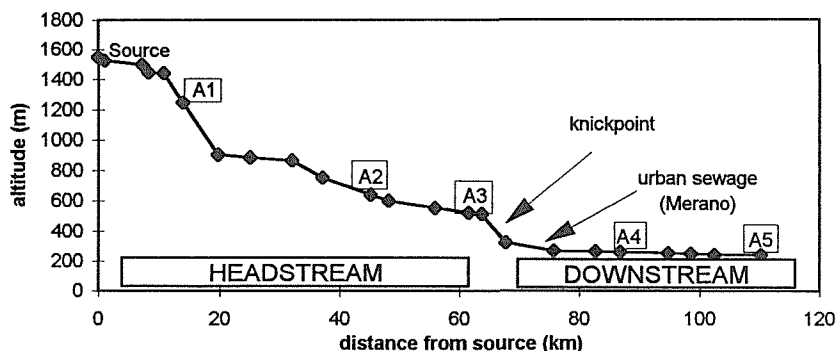


Fig. 1 Longitudinal Section of the Adige River, showing the locations of the five sampling stations (A1, A2, A3, A4, A5), of the knickpoint and the urban sewage effluent of the city of Merano.

Field analysis

Macroinvertebrates were sampled using a modified Surber-Sampler for two years during spring, summer and autumn. Three replicates were collected at each sampling site and individuals were classified taxonomically and associated to the respective functional feeding groups (Anderson & Sedell, 1979; Wallance & Merritt, 1980) to evaluate variety of taxa, the relative abundance of each taxon and macrobenthic total density. During every sampling period, hydrological measurements (current velocity, depth, river width) were performed simultaneously with chemico-physical analyses: total phosphorus (P_{tot}), ammonia (NH₄), chemical oxygen demand (COD), biological oxygen demand (BOD₅).

To determine the grain-size composition of the sediment, samples of substrate were fractionated through a stock of sieves with mesh size following a geometric series (Wentworth scale). The dry weight of each fraction and the cumulative percentage were measured. The amount of organic matter, subdivided into FPOM and CPOM in relation to the particles size (FPOM < 1 mm > CPOM), was determined by the ash-free dry weight evaluation (dry weight 70°C for 72 h; ash weight 500°C for 5 h).

Statistical analysis

To evaluate the similarity between different macroinvertebrate associations, Multidimensional Scaling was applied. This multivariate analysis graphically represents the similarity relationships among the samples collected in each season in the five stations investigated. The matrix utilized for the comparison was set by considering the entire macrozoobenthic community. The matrix was reduced in size by the exclusion of all taxa with abundance lower than 5% of the total number of individuals in each sample to rule out rare taxa, and data were double square root transformed to reduce the weight of abundant species. The matrix from which the final configuration was built up was a Q-matrix using the dissimilarity index "City-Block, distance of Manhattan" (Field *et al.*, 1982).

All the comparisons in macroinvertebrate data recording, were tested by the analysis of variance (one way ANOVA).

RESULTS AND DISCUSSION

The general characteristics of each site are summarized in Table 1. The sampling sites are located at progressive distances from the source. Chemically, site A4 shows the highest values of BOD₅, COD, P_{tot} and NH₄ concentrations. This could be attributed to the untreated sewage, which reduces the water quality of the second river section.

The grain-size distribution of sediment samples does not change seasonally. Figure 2 represents the mean annual values in the five stations. There is a clear separation between the stations located in the upper section and the others in the lower section. The trend obtained from cumulative values of the dimensional

Table 1 Abiotic characterization of the sampling stations.

Site	A1	A2	A3	A4	A5
Altitude (m)	1000	700	380	310	260
Distance from source (km)	10	50	70	90	120
Width (m)	2	8	10	20	60
Current velocity (m s^{-1})	1.5	1.1	1.3	0.8	0.9
COD	2.24	1.85	2.37	4.45	3.65
BOD ₅	3.04	3.12	3.61	5.24	3.18
P _{tot} (mg l^{-1})	86.56	58.76	94.92	178.11	117.06
NH ₄ (mg l^{-1})	68.94	37.98	77.15	135.92	82.26

sediment fractions shows that the upper stations (A1, A2, A3) are characterized by high portions of coarse-sediment fractions, whereas the lower stations (A4, A5) are characterized by a fine-sediment layer almost devoid of coarse particles. The decreased substrate heterogeneity and the prevalence of fine structured sediments reduce the variety of microhabitats for the macrobenthic fauna (Cummins & Lauff, 1969; Clup *et al.*, 1983).

In the lower river section, the high amount of fine sediment can be correlated with the absence of sharp slope variations. The critical velocity required to transport the particles is not reached and the accumulation of fine sediments is favoured.

Figure 3 illustrates the average seasonal abundance of the organic matter in all stations, expressed as CPOM and FPOM. Again, only the first three stations differ significantly from the last two. This shows a different trophic characterization of the two river sections: in the upper part the organic matter is mainly represented by CPOM, while in the lower part CPOM is reduced to less than 5% of the total amount of organic matter. The high values of FPOM down from the knickpoint are most likely increased by the organic input of urban sewage. The low amount of CPOM in the second river section could not be attributed to the reduction of potential input from the riparian zone, because these areas are comparable in both sections. We suppose that the rectification of the second section and the clogging of the substrate layer (due to the high input of urban pollution) reduce the retentive capacity of the stream and decrease the CPOM trapping efficiency. In the upper section, the CPOM

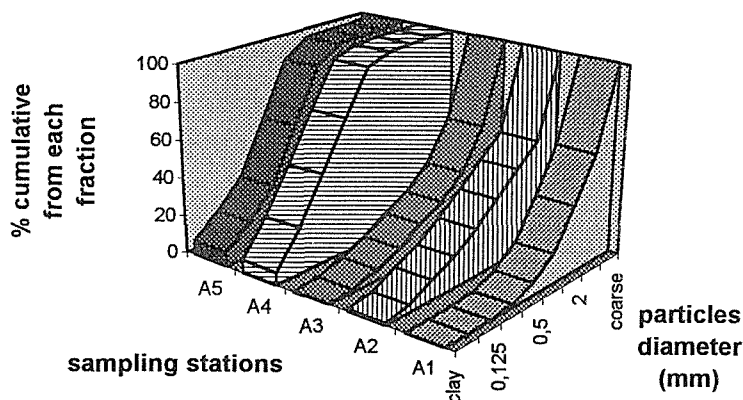


Fig. 2 The grain-size distribution of the substrate overlayer expressed as cumulative values of each sediment fraction and for each sampling station.

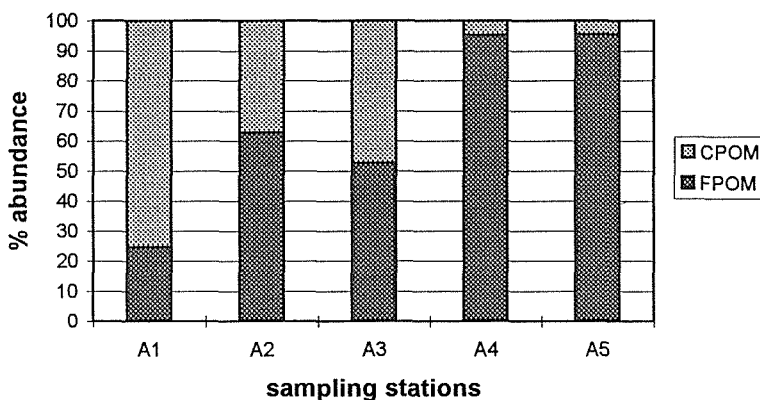


Fig. 3 Abundance (as percentage) of CPOM and FPOM found in every sampling station.

can be stored in several ways (high stream-bed roughness, stones, debris dams, gravel bars) (Naiman, 1982; Clup *et al.*, 1983), while in the lower section the absence of retentive structures does not allow CPOM deposition.

The variety of taxa, the relative abundance of each taxon, and the total density during the same season of the two years investigated were analysed by one way ANOVA. The results did not show significant differences, therefore the seasonal data of different years were pooled.

Table 2 lists the taxa with mean seasonal abundance greater than 5% during at least one season. Some taxa were present along the entire investigated tract of the Adige River, even if their abundance changed among the sites. For example, Trichoptera of the genera *Mesophylax* and *Metanoea* were one of the dominant taxa in the upper part of the river, but colonized also the lower part, although with a much lower density. Chironomidae and Tubificidae indeed were present in the source area, but became abundant only in the downstream zone. Furthermore, the macroinvertebrate community was characterized by different taxa in the upper and the lower section of the Adige River. In the upper section (especially in the station A1), numerous taxa of Plecoptera, Ephemeroptera, Diptera and Trichoptera could be found, whereas the lower section was dominated by Oligochaeta, Hirudinea, Nematoda, Chironomidae and Tubificidae. The first station (A1) had the highest number of taxa during all seasons (Table 3). The number of taxa gradually decreased in the upper river section, from station A2 to station A3. The lowest number of taxa during every season appeared in station A4, located below the urban sewage effluent. The low water quality (dissolved oxygen concentrations, biological oxygen demand, suspended organic compounds) (Table 1) in this river tract could be the main reason for the increasing densities of Tubificidae and Chironomidae (mostly *Tubifex tubifex* and *Chironomus thummi*).

The highest values of total macrobenthic densities in station A1 evidence the dissimilarity existing between this station and the others during every season. In every station, the total densities of the spring samples are significantly higher than in any other season.

Multidimensional scaling, applied to the entire community structure, shows that in all seasons station A1 had a higher dissimilarity of the macrobenthic communities

Table 2 Presence and abundance of the dominant taxa (taxa reaching a cumulative abundance of 95%). Fg = functional group.

Taxa	Fg	A1	A2	A3	A4	A5
<i>Leuctra nigra</i>	SH	***				
<i>Baetis alpinus</i>	GC	***				
Psycodidae	SC	**				
<i>Amphinemura standfussi</i>	SH	***				
<i>Dinocras cephalotes</i>	PR	***				
<i>Epeorus sylvicola</i>	SC	**				
<i>Hydropsyche pellucidula</i>	FF	**				
Limoniidae	PR	**	*			
Tipulidae	SH	**	*			
Brachycentridae	SH	*	*			
<i>Rhitrogena semicolorata</i>	SC	*	*			
<i>Drusus sp.</i>	SH		**	*		
<i>Rhyacophyla dorsalis</i>	PR		*			
<i>Bithynia tentaculata</i>	SC		*			
<i>Mesophylax sp.</i>	SH			*		
<i>Ecdyonurus venosus</i>	SC	**		*		
Simuliidae	FF	***	**	*		
<i>Metanoea sp.</i>	SH	***	**	*		
<i>Ephemerella ignita</i>	SH	***	*	*		*
<i>Baetis rhodani</i>	GC	***	***	***	*	**
Chironomidae	GC	*	*	**	***	***
<i>Leuctra fusca</i>	SH	*	**	**	*	*
Tubificidae	GC		*	*	***	***
Lumbriculidae	GC				***	***
Naididae	GC			*		**
Hydracarina	PR		*			*

* (1 – 2.5%),

** (2.5 – 5%),

*** (> 5%).

Table 3 Total macrobenthic densities (number of individuals m⁻²) and number of taxa.

Site	A1		A2		A3		A4		A5	
Summer	1559.17	27	760.83	20	1057.50	17	849.17	14	806.67	13
Autumn	2788.33	29	1438.33	22	510.00	18	1165.83	11	350.00	14
Spring	6714.17	24	4683.33	16	3025.83	16	8305.00	11	5345.83	13

in comparison to the others (Fig. 4). The differences among the stations seem to decrease with the increasing distance from the source. Indeed, Fig. 4 shows a higher similarity in community structures in the stations from A2 to A5 and also the separation of the samples collected in spring. Surprisingly, station A3 and A4, one located before and one after the knickpoint, do not show great dissimilarities in terms of taxonomic composition. Taxonomically the macrobenthic communities change downstream, but the differences are gradual and the knickpoint does not induce an abrupt interruption in the biotic continuum.

A macroinvertebrate discontinuity between the two river sections becomes evident from an analysis of the functional composition of the communities. Data

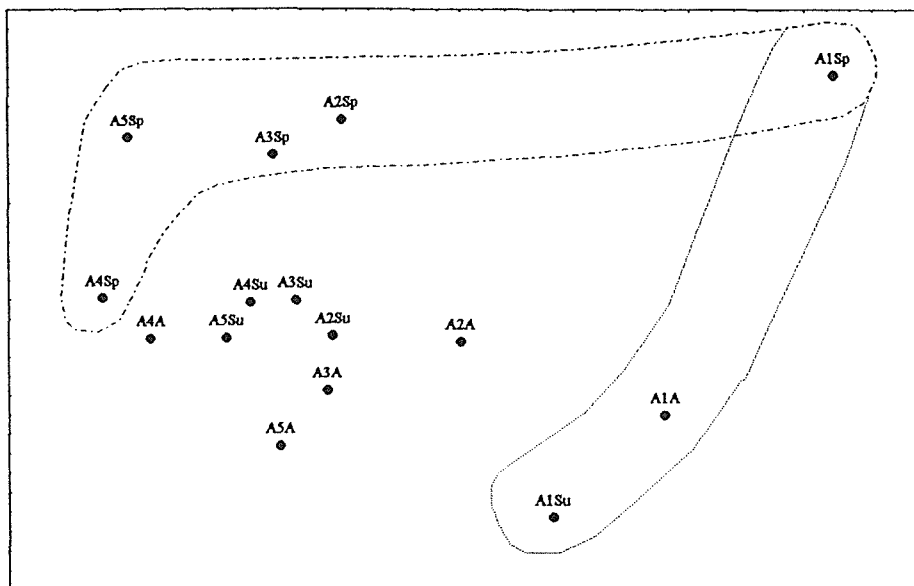


Fig. 4 Final configuration of the multidimensional scaling (A1–A5 Su = summer samples; A1–A5 A = autumn samples; A1–A2 Sp = spring samples).

reported in Fig. 5 show the relative abundance of functional feeding groups. Detritivorous abundance (shredders plus scrapers, SH+SC), slowly decreases from the first to the third station. After the knickpoint their presence is drastically reduced. The opposite trend is revealed by the relative abundance of collectors (gathering collectors plus filter feeders (GC+FF), which are more abundant after the knickpoint. The abundance of detritivorous and collectors, compared for each season by the analysis of variance (one way ANOVA), does not show significant differences except for stations A4 and A5 in every season. However, comparing all stations in all seasons, a significant difference was always found between every station above the knickpoint and everyone after it. This underlines the importance of the knickpoint in the characterization of the two main sections of the Adige River. The strong

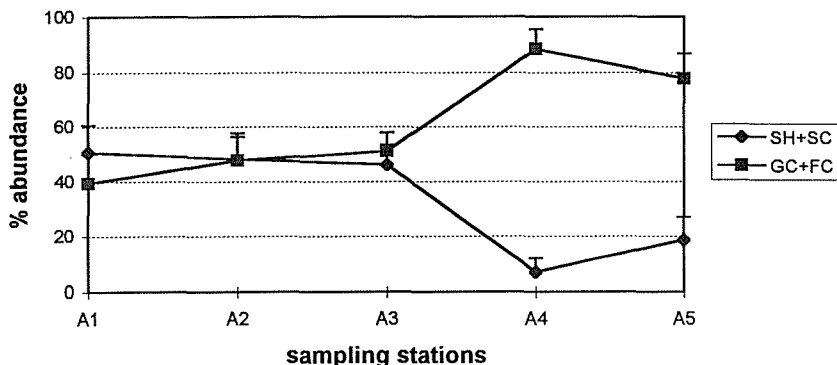


Fig. 5 Mean annual abundance (as percentage) of functional feeding groups (SH = Shredders; SC = Scrapers; FF = Filter Feeders; GC = Gathering Collectors).

prevalence of collectors over detritivorous in the lower river section can be related to the fine particulate substrate and the massive nutrient load.

CONCLUSIONS

The upper and the lower section of the Adige River differ in substrate heterogeneity and food resource. The absence of slope variations in the second river section and the human impacts (rectification of the river margins and the untreated sewage effluent) reduce the amount of coarse sediment particles and CPOM favouring the formation of a fine substrate layer and deposition of FPOM. These differences between the two river parts result in an interruption of the biotic continuum. The sharp biotic discontinuity becomes more evident considering the functional approach rather than the taxonomic analysis: in fact, the taxonomic composition changes gradually proceeding downstream, while the shredders and deposit feeders dominate respectively in the upper and lower river section. The knickpoint and the urban sewage overloading may act together in promoting a biotic interruption which apparently seems to divide the stream into two different faunistic compositions.

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Influence of surface and subsurface flow on distributions of particulate organic matter and inorganic fine particles in perialpine stream sediments

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Abstract The relevance of vertical hydrological exchange processes for the retention of particulate organic carbon (POC) in channel sediments was assessed by comparing the depth distributions within three types of interstitial flow direction; infiltration, exfiltration, and horizontal advection. The POC content increases down to a depth of 50 cm in all three hydrological exchange types. However, the differences between 50 and 100 cm are only significant in the region of exfiltrating groundwater. Inorganic fines tend to increase down to depths of 150 cm, with the exception of the exfiltration type, where the highest inorganic fine contents are found at a depth of 50 cm, indicating an inhibiting effect of the net upward movement of groundwater. The POC/TP ratio, defined as the ratio between POC and the total of all particles, decrease with depth for all types of interstitial flow direction. In the exfiltration type the POC/TP ratios are lower than in the other two types down to a depth of 100 cm. Frequent flood events cause high temporal variability of interstitial particle parameters.

INTRODUCTION

Particulate organic carbon (POC) is the food resource for most animals living in streams. Many temperate fluvial ecosystems depend on the mainly seasonal inputs of allochthonous POC from riparian zones (Bretschko & Moser, 1993). However, streams are transport systems and thus the retention of POC is relevant for the year-round availability of POC to the biota (Speaker *et al.*, 1984). The channel sediments form a major storage site due to their porosity and increasing stability with sediment depth (Naegeli, 1997). POC is imported into the sediments during bed load moving spates (Bretschko, 1991) and by depth filtration, i.e. the transport and deposition of particles within interstices, depending on interstitial currents and pore sizes (Brunke & Gonser, 1997a). Therefore the import is restricted by the degree of consolidation of the channel bed and also by the permeability of deeper strata. These properties of the sediment structure are markedly influenced by the proportion of inorganic matrix fines (<1 mm). Excessive amounts of inorganic particles (IPs) may lead to a clogging of the top layer of the channel sediments (colmation). They can also fill the interstitial voids, thereby reducing the available subsurface habitat (Brunke &

Gonser, 1997b). However, the deposition of IPs increases the surface area which can be covered by a biofilm in a given sediment volume. Leichtfried (1988) found high correlations between IPs and the POC content in the bed sediments of a calcareous mountain brook; most of the POC originated from the biofilm. Consequently, studies on the POC within the interstices must integrate the spatio-temporal dynamics of the IPs. In our study we investigated the role of the vertical hydrologic exchange (infiltration, horizontal advection, exfiltration), the surface flow regime and season in the distribution patterns of these particle parameters within the interstices.

STUDY SITE AND METHODS

The River Töss is a third-order calcareous gravel-bed river in northern Switzerland. At the study site (460 m a.m.s.l.) hydraulic gradients indicated groundwater exfiltration on the left-hand side of the stream, whereas river water infiltrates to the groundwater on the right-hand side. In midstream subsurface flow in the streamflow direction prevails (horizontal advection). Along the margins of the stream, where the infiltration and exfiltration zones are located, the uppermost channel sediments form an erosion-resistant armour layer, whereas in midstream, frequent bed load movement creates permeable open-framework gravel. The surface sediments had a mean grain size of 31 mm (± 26 SD) and the subsurface sediments, 11 mm (± 0.7 SD), with a mean porosity of 18.8% (± 6.6 SD). In each of the three zones steel piezometer pipes (internal diameter = 5 cm) were installed to four sampling depths (20, 50, 100 and 150 cm). Interstitial water containing the particles was collected by using an electric pump. Sampling was conducted at approximately monthly intervals between April 1995 and November 1996. For the analysis of the particles the samples were filtered through Whatman GF/F filters. The filters were dried at 60°C to determine total dry weight of the particles. Inorganic carbonate was removed from the filters with 3 M HCl. POC and PN were measured by a CHNS analyser (Elementar vario EL). POC and PN contents are referred to as the amounts of particulate organic carbon and nitrogen in 1 litre of sampled interstitial water, whereas POC/TP and PN/TP ratios are defined as the amounts of particulate carbon and nitrogen in mg g⁻¹ of total particle dry weight (*sensu* "POC concentrations", Leichtfried, 1988). The IPs were smaller than 1 mm and the size frequency distribution had a mode of 0.1 mm. The effect of the hydrological exchange type, depth and season on the particle parameters were analysed by the Kruskal-Wallis test.

RESULTS

Depth distribution of particles in different hydrological exchange zones

Overall, inorganic particles (IPs) within the interstices tended to increase with depth ($p < 0.001$). In the horizontal advection zone (HZ), the IPs increased gradually to a depth of 150 cm, whereas in the infiltration zone (IZ), the change with depth was discontinuous (Fig. 1). However, for both zones, only the IPs of the 20-cm and

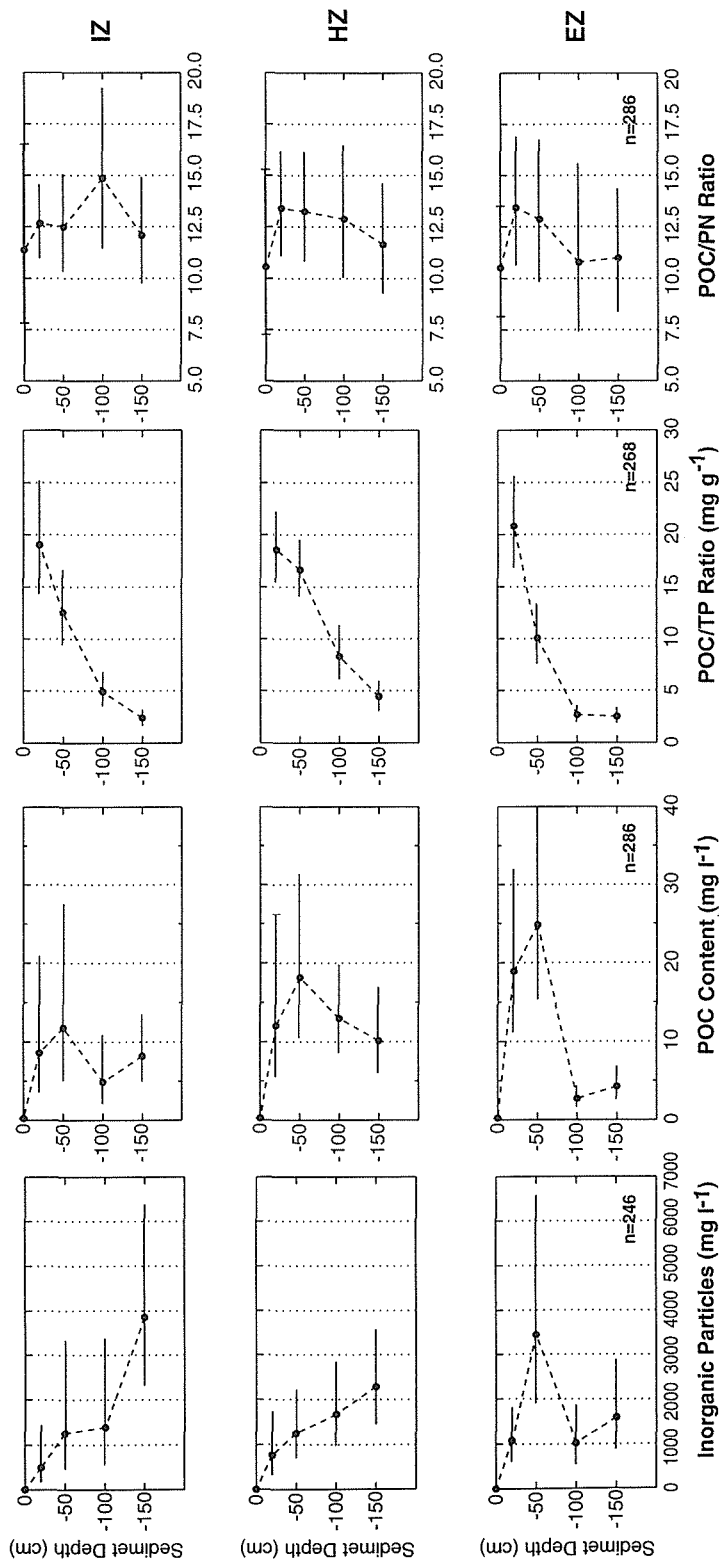


Fig. 1 Depth profiles of the geometric means of inorganic particles (IPs) (mg l^{-1}), the POC content (mg g^{-1}), the POC/TP ratio (mg g^{-1}) and the POC/PN ratio. Error bars: 95% confidence intervals, calculated after logarithmic transformation.

150-cm depth strata were significantly different for the IZ as a result of a high temporal variability. In the exfiltration zone (EZ), the IPs of the 50-cm depth stratum were significantly higher than in the 20- and 100-cm depth strata in spite of large confidence limits.

In all zones the POC content increased to a depth of 50 cm and decreased in the deeper strata ($p < 0.001$). However, the difference between the upper and deeper strata was significant only for the EZ. The POC content was significantly correlated to the amount of IPs ($r = 0.88$, $p < 0.001$, $n = 246$).

The POC/TP ratios differed significantly between the zones ($p = 0.005$) and decreased significantly with depth ($p < 0.001$). In the IZ and HZ the POC/TP ratios did not differ significantly from the 20- to the 50-cm depth stratum, whereas they differed significantly between 50- and 100-cm, as well as between the 100- and 150-cm depths. In contrast, the concentrations in the EZ, differed between the 20- and 50-cm depths and were similar in the depths of 100- and 150-cm.

The PN content was highly correlated with the POC content ($r = 0.96$, $p < 0.001$, $n = 286$); the POC/PN ratio did not vary significantly, neither with depth nor between the zones.

Distribution of particles with depth and season

The flow regime differed between the four seasons (Fig. 2(a)). In summer and autumn there was a moderate flood frequency and no major floods, while winter was characterized by one major flood and longer periods of low discharge; during spring several small and major floods led to a high flood frequency (Figs 2(a) and 3).

The interstitial POC contents and POC/TP ratios varied significantly among seasons ($p = 0.002$ and $p < 0.001$, respectively) (Fig. 2(b),(c)). The pattern of depth distribution of the POC content was similar in spring, summer and autumn, with higher contents at a depth of 50 cm than at 20 cm. The POC contents were the lowest in spring and differed significantly from POC contents in autumn and winter. This might be a result of the high flood frequency in spring relative to the other seasons. The POC contents were highest in winter, especially at a depth of 20 cm. However, the spatio-temporal variability, even down to depths of 100 cm, was very high, as indicated by the large confidence intervals (Fig. 2(b)).

The POC/TP ratios decreased with depth in all seasons. Except for winter the gradients between adjacent depth strata were steep. The POC/TP ratios were significantly higher in autumn compared to the other seasons (Fig. 2(c)). This is probably due to a higher import of POC following leaf abscission. In contrast, the high POC contents in winter were not reflected by high POC/TP ratios, thus indicating a high intrusion of IPs into the interstices.

A major flood and inorganic matrix fines

The effect of a major flood on the interstitial IPs can be shown by the samples taken before and after the most severe spate mobilizing the stream-bed sediments of the whole channel (Fig. 3). The IPs were highest at the margins of the stream after a

short time of low flow on 9 May. During the falling limb of a small spate on 16 May, the IPs were flushed out except those in the IZ at a depth of 100 cm. After the major flood on 1 June we could not access the stream until 19 June due to the

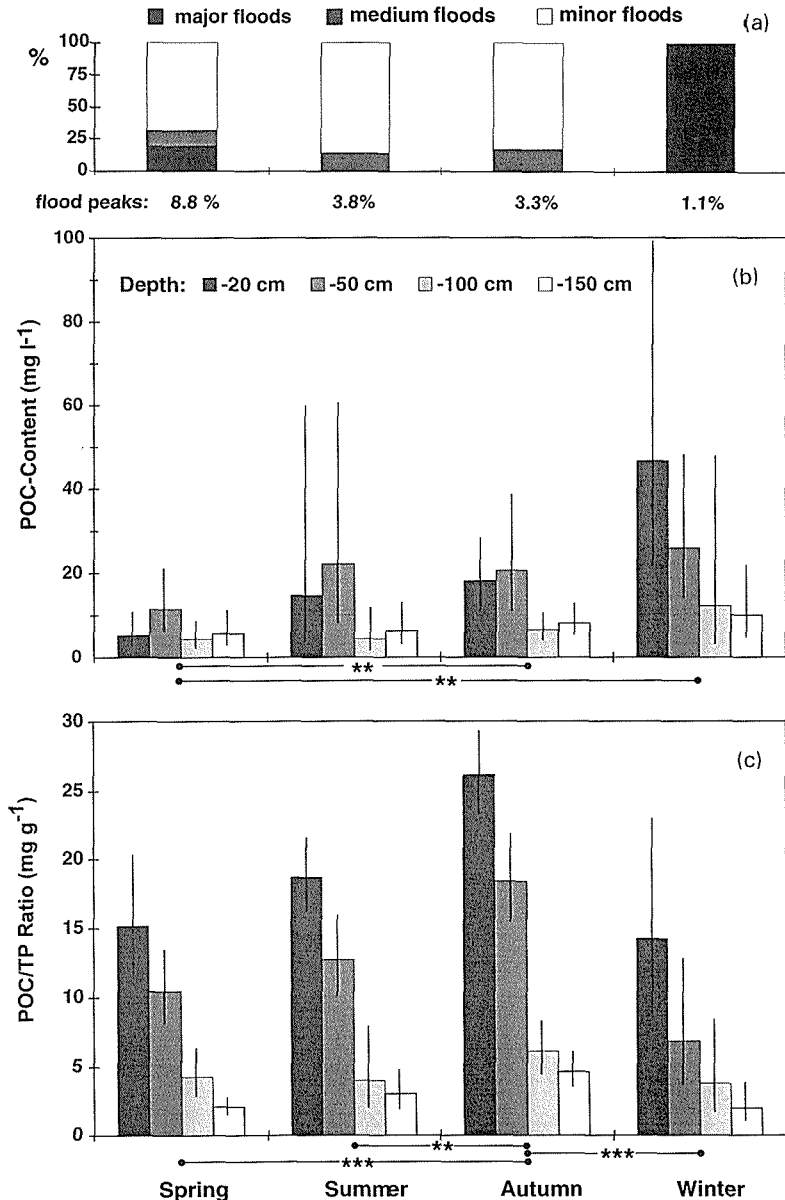


Fig. 2 Seasonal variations of the flood regime (a), the geometric means of the POC content (b) and POC/TP ratio (c) with depth. The flood peaks indicate the percentage days with flood peaks relative to all days of a season. The bars indicate the relative portion of flood types in the percentage of flood peaks; major floods, $Q > 80$; medium floods $Q = 40-80$; minor floods $Q = 20-40$ m³ s⁻¹. Asterisks indicate significant differences between the seasons for the POC; ** = $p < 0.01$; *** = $p < 0.001$. Error bars indicate 95% confidence intervals, calculated after logarithmic transformation.

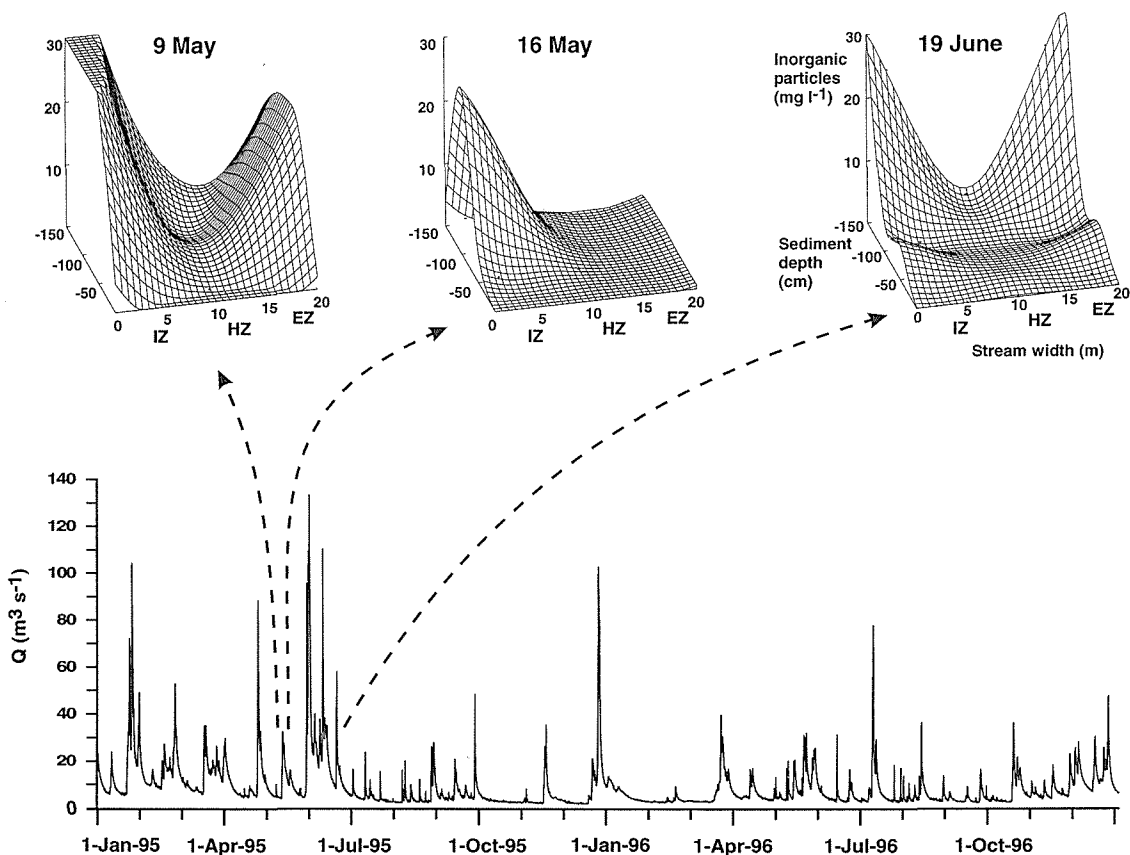


Fig. 3 Discharge regime of the River Töss during the sampling period and the amount of inorganic particles (IPs, mg l^{-1}) within the interstices of the infiltration zone (IZ), horizontal advection zone (HZ) and exfiltration zone (EZ) at sediment depths of 20, 50, 100 and 150 cm and in the surface water on 9 May, 16 May and 19 June 1995.

high water level. On that particular day the IPs were low, except for higher contents at depths of 150 cm in the margins.

DISCUSSION

The standing stock of interstitial POC is maintained by the import from the surface habitat, as well as by the microbial conversion of dissolved organic carbon to biofilms attached to solids. The general spatio-temporal patterns of POC storage in the sediments of the River Töss result from a complex of interacting superimposed factors: sediment depth and structure, vertical hydrological exchange, surface flow regime and season. These factors control the loading of particles entering the interstices (Walling, 1996). In addition, interstitial currents determine deposition and mobilization within voids and the straining by pore sizes (Herzig *et al.*, 1970). The content of inorganic matrix fines reflects the abiotic transport of particles. They accumulate with depth due to their density. This pattern was also found for other gravel-bed streams (Eglin, 1990; Panek, 1994). However, the upward hydraulic

forces of upwelling groundwater may prevent the intrusion of particles into deeper sediment layers. This can result in temporary accumulations in intermediate depths or in a release of particles to the surface water when thresholds of interstitial currents are exceeded (Schälchli, 1993). We measured an export of particles in a spring at the stream's left margin in the range of 130–218 mg l⁻¹. Floods may also induce a flushing effect on matrix fines by a release of fines into the stream during bed load movement or by transport into the aquifer by bank infiltration.

The interstitial POC content is correlated to the IPs, although this is not reflected by the general vertical patterns below a 50-cm depth, except in the EZ. There, the low POC contents are a consequence of the upwelling groundwater. In the IZ and HZ the decline of POC contents below a 50-cm depth may reflect a synergetic effect of biotic degradation, a reduced import due to a decreasing sediment permeability by increasing amounts of fines and the increasing distances from the fluctuating surface flow, which dynamically influences interstitial throughflow. The variability in the POC content reveals the effect of discontinuous POC import. However, particle transport can be assumed to be a continual process along interstitial flow paths.

The flashy discharge regime of the Töss may be responsible for the spatio-temporal variability of interstitial POC content. In spring, flood peaks were the most numerous and varied in intensity. Concomitantly, the interstitial POC contents were the lowest. In contrast, the discharge in winter was dominated by low flow conditions and POC contents were correspondingly high. These high contents cannot be due to the import from riparian zones because the POC/TP ratios were significantly higher in autumn, demonstrating the importance of deciduous leaf litter input during that season.

The hydrograph peaks were artificially divided into minor, medium and major floods, since it was not possible to find discharge thresholds for initial motion of the bed load. This can be due to the flood history and different degrees of consolidation of the channel bed (Reid *et al.*, 1985). The midstream sediments were moved most frequently, depositing a permeable open-framework gravel, which facilitates the interstitial transport of particles. When such unconsolidated sediments are filled with fines, the deposition process begins in deeper layers and proceeds continuously to upper layers (Einstein, 1968). In agreement with these sediment properties we found higher amounts of pure organic particles in the HZ, which explain the higher POC/TP ratios in the HZ. These organic particles—originating from plants (personal observation)—cannot be buried during floods, because bed load movement to this depth would have removed our piezometers.

Flood events, vertical exchange of waters, and the channel sediment structure are important determinants of the distribution of inorganic fines and POC at depths considerably more than 50 cm below the stream bed. Thus transport and deposition of fine particles within the interstices (depth filtration) are not restricted to the upper layers of channel sediments in gravel-bed streams, where they may induce clogging of the interstices (colmatation). Rather they reach further down and influence the habitat characteristics of deeper sediment layers.

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Using spring biota of pristine mountain areas for long-term monitoring

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Abstract Thirty springs in the Adamello-Brenta Regional Park (Trentino, Italy) distributed over a large altitudinal range (from 1073 to 2172 m a.s.l.) were studied between 1992 and 1996. A high diversity of plant and animal communities including algae (mainly cyanophytes and diatoms), mosses and higher plants, macroinvertebrates (the main groups being midges, stoneflies, caddisflies), water mites, microcrustaceans, and crenobiontic snails was identified. The springs situated in pristine areas allow the detection of slight shifts as well as long-term effects of impacts on their drainage basins much more clearly than freshwater environments altered by human activity. In this paper we discuss the use of spring biota for the definition of the trophic status, ongoing acidification processes, and organic pollution. After a broad screening of the biota within a basic investigation, continuous survey of water mites is recommended for checking the persistence of stable conditions and the survey of diatoms for monitoring acidification and eutrophication processes.

INTRODUCTION

Pristine environments in remote areas are suitable sites for the detection of environmental changes. This is particularly true for springs because of their characteristic physico-chemical stability. The specific features of these environments have caused the evolution of crenobiontic (living only in springs), crenophilous (living mainly in springs), and crenoxenic (living occasionally in springs) organisms.

Most attempts to use spring organisms for monitoring pollution by organic substances and distinguish stable springs from temporary waters were essentially concentrated on macroinvertebrates (Williams, 1991; Bodon & Geiter, 1995). The present study uses for the first time a broad spectrum of organisms to detect environmental changes in order to develop a long-term monitoring programme, i.e. detection of changes and/or trends in biotope/water quality over longer time periods independent from inter-annual variability.

Background information about the case study

From 1992 to 1996, 30 springs from all major valleys in the Adamello-Brenta Regional Park (618 km²) have been studied. The Adamello mountain group consists of igneous rocks, such as granites, whereas the Brenta group is mostly composed of limestones and dolomites. The springs, distributed over a large altitudinal range

(from 1073 to 2172 m a.s.l.), are characterized by low to medium discharge (mean seasonal discharge from 16 ml s⁻¹ to 80 l s⁻¹) and by a wide range of substrate sediments (from silt to boulders). The annual coefficient of variation ($C_v = s/\text{mean}$; Gordon *et al.*, 1992) is between 0.6 and 1 in 15 springs, between 0 and 0.5 in nine, and between 1.2 and 1.8 in six. This last group includes four springs, all in the Brenta group, which tend to dry out in the winter. The temperature fluctuation throughout the year is from 0.3 to 1.1°C in 16 springs, from 3.1 to 6.4°C in ten and from 1.5 to 2.5°C in four.

Most of the springs studied are of the rheocrene type (flowing springs). In addition two hygropetric rheocrenes (small springs seeping out from the rock), two helocrenes (seepage springs), one rheohelocrene (a seepage source emerging on a steep slope), and one limnocrene (a pool spring) have been identified. Helocrenes differ from rheocrenes because of their oxygen undersaturation (30–80%; rheocrenes: 76–102%), higher carbon dioxide content (15.0–31.0 mg l⁻¹; rheocrenes:

Table 1 Spring biota (upper part autotrophes, lower part heterotrophes), study methods, and relevant characteristics for long-term monitoring based on a study in the Adamello-Brenta Regional Park. When not otherwise specified information about methods and rare species is available in Cantonati (1998).

Autotrophes	Higher plants	Mosses	Blue-green algae	Diatoms
Method	Pattern mapping and vegetation analysis		Charting and sampling of all coverages	Separate sampling of all substrates
Taxa number total	245	58	40	250
Red list taxa or rare spp. (source)	27 (Niklfeld, 1986)	19 (Niklfeld, 1986)	10 (Rott <i>et al.</i> , 1998)	120 (Lange-Bertalot & Steindorf, 1996)
Mobility	Any	Any	Very reduced	Reduced
Dispersal	Effective	Difficult	Effective	Very effective
Distribution	Wide	Restricted	Wide	Very wide
Crenobionts	Any	Any (?)	Any (?)	Any (?)
Crenophiles	Some	Several	Some (?)	Some (?)
Inferential impacts	Flow stability	Flow stability	Flow stability organic pollution metals eutrophication	Flow stability organic pollution metals eutrophication acidification
Heterotrophes	Copepodes and ostracodes	Water mites	Crenobiontic snails (Hydrobioidea)	Other invertebrates
Method	100 µm mesh net considering all sub.	100 µm mesh net considering all sub.	Sorting from dried and sieved sediments	0.33 mm mesh net kick samples
Taxa number total	21	19	1	96
Red list taxa or rare spp. (source)	7	10	1 (CNR, 1980–1985)	12 (CNR, 1980–1985)
Mobility	Medium	High	Little	High
Dispersal	Difficult	Medium	Very difficult	Medium
Distribution	Defined	Defined	Very defined	Defined
Crenobionts	Some	Many	All	Any (?)
Crenophiles	Many	Some	Any	Some
Inferential impacts	Flow stability organic pollution	Flow stability organic pollution metals (?)	Organic pollution	Flow stability organic pollution

1.5–19.0 mg l⁻¹), and lower pH (5.8–6.4; rheocrenes on siliceous substrate: 6.0–7.5). Spring waters from the Brenta exhibit a medium mineralization level (152–300 $\mu\text{S}_{25^\circ\text{C}} \text{ cm}^{-1}$), whereas those from the Adamello (11–64 $\mu\text{S}_{25^\circ\text{C}} \text{ cm}^{-1}$) are little mineralized (Cantonati, in press). Some of the latter springs have extremely low alkalinity and, therefore, are endangered by the effects of acid rain and snow. Rainfall may play a major role in determining nitrate values (74–1010 $\mu\text{g l}^{-1} \text{ NO}_3\text{-N}$), which are commonly higher than expected for pristine environments; however, the total phosphorus values (mean = 6.6 $\mu\text{g l}^{-1}$; s = 3.6 $\mu\text{g l}^{-1}$; n = 75) allow us to classify all sites as oligotrophic. The majority of springs can be considered pristine. However three springs are affected by nutrient input from pastures, three are intercepted for drinking purposes, one has been transformed into a fountain and one is influenced by a nearby old iron and copper pyrite mine.

METHODS

Between 1992 and 1996 30 springs were repeatedly sampled. All data referred to in this paper were obtained through sampling that part of the spring stream where temperature did not rise more than 0.1–0.5°C in summer. By means of cooperation with specialists all the groups of organisms listed in Table 1 were studied in 19 springs of the central basin (River Sarca) of the Regional Park. In 11 springs in the other main basins (rivers Noce and Chiese), considered to cover the whole territory of the protected area, diatoms, water mites, microcrustaceans, and snails were analysed. The methods used are listed in Table 1. All specimens are stored at the Museo Tridentino di Scienze Naturali (Trento, Italy).

RESULTS AND DISCUSSION

The results show general characteristics and/or trends of mountain spring biota, which are reconsidered with respect to their possible use for long-term monitoring. Detailed information about all the plant and animal communities discussed is available in Cantonati (1998).

The macroscopic aspect of the springs is primarily determined by algae (mainly blue-green algae and diatoms, and, in fast flowing springs, the chrysophyte *Hydrurus foetidus*) mosses, and higher vegetation and shows in most cases a mosaic pattern formed by a dominant moss layer with interspersed higher plants. Mosses include submerged species (hydrophilous mosses, mainly liverworts) and/or species only partially soaked with water (hygrophilous mosses). Also crustous aquatic lichens are important, in particular *Verrucaria* spp. Among the higher plants, some are typical of places with elevated and constant moisture and others occasionally invade the spring area from the surrounding vegetation. On the lower side of the stones, within the mosses and the debris, several invertebrates can be found, including insect larvae (in particular midges, stoneflies and caddisflies), water mites, microcrustaceans (mainly copepods and ostracodes) and snails (Fig. 1).

Considering the limited surface area of these springs, a remarkably high number of taxa was found. In those 19 springs where all groups have been examined

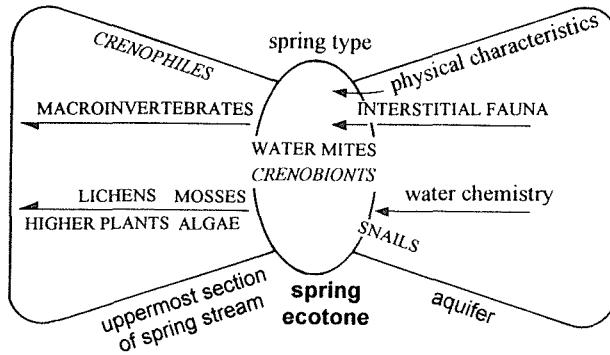


Fig. 1 Model illustrating the role of springs as ecotones linking the aquifer to the uppermost section of the spring stream; major organism groups and some important variables are indicated.

(Table 1), the total number of taxa ranged from 50 (in the spring transformed into a fountain and in the hydropetric rheocrene) to 113 (in one of the helocrene springs). Diatoms may be considered for a comparison between permanent and seasonally dry springs: the number of species ranged from 9 to 11 in the seasonally dry springs and from 9 (in a spring showing a slight heavy metal enrichment) to 54 (in the helocrene) in the permanent springs. With respect to biodiversity conservation and nature management it must be remembered that many rare and red list species (Table 1) were identified. Even taxa new for science were described from the material collected (Cantonati, 1998).

The only group that was found to be very rich in crenobionts is the water mites (more than 80% of the species found; Table 1). Most of the other invertebrate groups seem to include only crenophilous organisms which are also found in the upper part of the spring stream (Fig. 1). Only one species belonging to the superfamily Hydrobioidea, which includes crenobiotic snails very sensitive to pollution, was found. The low number of species and the restricted geographic distribution allows the use of these snails as monitors only in stations for which time series data are available. In contrast, autotrophic organisms do not seem to include crenobiotic species. However a limited amount of the species found can be defined as crenophilous.

Springs are ecotones (Fig. 1) linking groundwaters to surface waters. Some animals found in springs may live in the upper part of the aquifer and are washed out, others move to the surface sediments from the interstitial habitat. At low altitudes, stygobionts (organisms living only in groundwater) are a common component of the spring fauna (e.g. Hynes, 1976). In our research, we did not find any stygobiotic organism, but several stygophiles, which by definition can also live in the spring boil. This is not due to the sampling methods that were not specially chosen for groundwater organisms, but to the difficulties of stygobionts to recolonize habitats only recently left free by the glaciers, due to their limited dispersal capabilities. Indeed, Crema *et al.* (1996), considering comparable Alpine springs with special reference to macroinvertebrates and interstitial fauna, found that stygobionts were poorly represented (only 4%).

Sediment quality and flow character are key variables when discussing the effect

of limnological spring type on the biota. In fact, one of the main differences between helocrenes and rheocrenes is the presence of silt and debris in the helocrenes and the dominance of coarser particle sizes (mainly pebbles, boulders and gravel) in the rheocrenes. This difference may be the main cause for the absence of cyanophytes in the helocrene springs, since an epilithic substrate is fundamental for blue-green algae. Furthermore, helocrenes differ from rheocrenes by the presence of limnophilous invertebrates such as cockles. The different structure of the substrates in the spring types is important also for the occurrence of microcrustaceans and water mites. The diatom associations were highly diverse in helocrene springs, showing a remarkable number of large species with well silicified valves and large variety of species of the genera *Eunotia* and *Pinnularia*. The presence of species with large frustules might be related to the higher silica content for helocrene and rheohelocrene springs ($5.5\text{--}8.0\text{ mg l}^{-1}$), whereas the high diversity of *Eunotia* and *Pinnularia* species can be linked to slightly acidic, but sufficiently buffered waters.

The results of multivariate analyses (TWINSPAN, CANOCO) showed that the following environmental variables are best correlated with species composition: pH for diatoms, conductivity for blue-green algae (Cantonati *et al.*, 1996), discharge for macroinvertebrates and flow rate as well as spring type which affect the moisture of the adjacent area for higher plants. Geochemical variables and the degree of shading are important regulating variables for the autotrophic organisms, whereas discharge has a more or less pronounced impact on all communities considered (Table 1). Marked discharge variations and, in particular, the lack of water for some time during the year, influence the composition of the biota: rare cyanophyte species and crenobiontic or crenophilous invertebrates only occur in permanent springs with stable discharge, whereas seasonally dry springs are characterized by a poor fauna and very low diverse diatom associations, with significant portion of aerophilous species. The persistence of the higher plant community pattern throughout the years indicates hydrological stability. According to van der Kamp (1995), persistence and variability of flow are the most important factors for spring organisms. Altitude was found to be important for the higher vegetation and, rather unexpected, for cyanophytes. Diatom communities are clearly influenced by pH (Table 1) and alkalinity and allowed us to identify springs endangered by acidification. On crystalline substrate, where very poorly mineralized waters are common, the availability of a biological method which allows the distinction of natural low alkalinity situations from those caused by acid depositions is of fundamental importance. Beierkuhnlein (1996) proposed to use springs on siliceous substrate for an early identification of acidification problems, because here acidification problems may be recognisable before damage in the forest is detectable. Moreover diatoms are suitable indicators not only for the monitoring of acid loads, but also for nutrient loads and hence for both of the two major actual impacts on freshwater environments (Psenner, 1994).

Organic pollution can also be detected by several other invertebrate groups including many sensitive taxa (e.g. midges, stoneflies and caddisflies). The nutrient status is best expressed by the algal associations (Table 1). Cyanophytes were present mainly with non-heterocytous forms, linked to an excess availability of nitrates (Cantonati *et al.*, 1996) and nutrients (in particular nitrates) were important regulating variables in determining the distribution of diatoms.

CONCLUSIONS

Our results show that permanent springs with small discharge fluctuations have to be selected for biological monitoring since they are characterized by stable biota, beyond marked differences from spring to spring.

Spring communities are highly diverse and include several suitable indicator groups which allow us to evaluate in particular runoff characteristics, acidification and eutrophication risks.

For long-term (detection of trends beside year to year variations) monitoring of the biota the following schedule is recommended:

- (a) Firstly, the initial broad screening which should include as many components of the biota as possible and document the biodiversity of springs. In particular, the investigation of crenobiontic water mites is necessary to select permanent springs.
- (b) Secondly, the monitoring of selected organism groups once a year and over longer time periods (at least 5–10 years). In particular, water mites need to be studied to check the persistence of environmental stability and diatoms are recommended to analyse acidification/eutrophication processes due to local and/or atmospheric inputs.

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High Alpine streams as models for ecological gradients

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Abstract Because of their location in the same climatic and geographical area, glacier- and spring-fed streams show similarities in their ecological conditions, but also pronounced differences in respect of physico-chemical characteristics. According to the steep-gradient nature of high Alpine areas, some of these variables alter at a small temporal and spatial scale, affecting species composition and abundance of the stream fauna. Comparing the glacier-fed Rotmoosbach and the spring-fed Königsbach in Ötztal (Tyrol) we evaluate community-organizing forces for the benthic fauna and detect adaptations in the occurring species. In our ongoing study we focus on the benthic biocoenosis in these two streams, primary physical and chemical variables that determine aquatic invertebrate distributions, trophic relationships in the aquatic communities, and colonization strategies of some of the dominant species in these streams.

INTRODUCTION

In areas of high altitude as well as high latitude, located between the permanent snow and the tree line, three types of stream ecosystems can generally be distinguished: glacier-fed (glacier-melt dominated, kryal), spring-fed (krenal) and rainfall, snow-melt dominated (rhithral) streams (Ward, 1994). Because of their location in the same climatic and geographical area, glacier- and spring-fed streams share a common suite of ecological conditions, but show some differences in physico-chemical respects (Steffan, 1971b; Ward, 1994). A comparison of both stream types enables the evaluation of community-organizing forces (runoff, turbidity, nutrient availability, etc.) for the benthic fauna and the detection of adaptations in the occurring species.

Although the hydrology and physico-chemical features of high altitude streams have been studied intensively, their ecology has received little attention. Essential matters regarding the importance of the physical and chemical environment in shaping succession and colonization strategies of the biota and/or possible adaptations of the organisms (single taxa or communities) have been disregarded so far. The role of allochthonous input, biomass transfers and energy flux between the aquatic and terrestrial ecosystems and food webs still remain unclear.

From a synthesis of the limited literature on the ecology of glacier-fed rivers, Milner & Petts (1994) proposed a general, qualitative model relating zoobenthic gradients to temperature and channel stability, as a function of distance from the glacier margin, and time since deglaciation. Steffan (1971a,b) characterized glacier

brooks by their extremely low summer temperature, strong fluctuations in discharge, and unstable substrate. In comparison with spring-fed streams in Alaska he found another important factor determining the biocoenosis of glacial streams. Low conductivity (usually $< 50 \mu\text{S cm}^{-1}$) might be the main limiting factor, which is besides substrate instability, suspended solids and water temperature, responsible for low primary production, which in turn, is the main cause for the low species number of this coenosis.

In our still ongoing comparison of high Alpine streams we:

- (a) investigate the benthic biocoenosis (aquatic invertebrates and algae) in a glacier-fed and a spring-fed stream,
- (b) determine the primary physical and chemical variables that determine aquatic invertebrate distributions in these streams,
- (c) apply our findings to the conceptual model of Milner & Petts (1994) concerning the longitudinal zonation of stream invertebrates in glacial streams,
- (d) study the longitudinal succession, site specific seasonal variation and colonization strategies of some of the dominant taxa,
- (e) describe the trophic relationships in the aquatic communities, and
- (f) characterize natural stream types which in turn can serve as ecosystem references for stream biological assessments and evaluation of ecosystem degradation.

In this paper we present our preliminary results after one year of fieldwork in high Alpine streams.

STUDY AREA

The study streams drain the northern slopes of the Austrian central Alps near Obergurgl, Ötztal (Fig. 1). The glacial stream flows through the Rotmoos Valley where the glacial retreat expands 300 m from the glacier snout to an area of approximately 3.5 km downstream. It represents a more or less flat area where distinct deglaciation zones can be distinguished:

- (a) the outer Rotmoos Valley for several centuries ice free,
- (b) end and side moraines from 1850,
- (c) end and side moraines from 1920, and
- (d) the recently ice free area (up to the glacier snout).

Like the glacier-fed Rotmoosbach, the spring-fed Königsbach drains a side valley of the Ötztaler Ache. Both streams run parallel; the Königsbach has no glacier in its drainage area, but about the same gradient and orientation.

METHODS

For an appropriate comparison four study sites per stream were chosen at about the same elevation (Fig. 1). In the glacial stream the four biggest tributaries serve as additional sites to evaluate their potential effect as pools for colonization and for altering the environmental conditions in the stream.

At each station the physical and chemical environment was monitored, using temperature-logging systems at seven stations, discharge measurements using a dip

stick, pH and conductivity reading at every sampling period and each station. Concentrations of organic and inorganic matter (suspended solids, AFDW) were

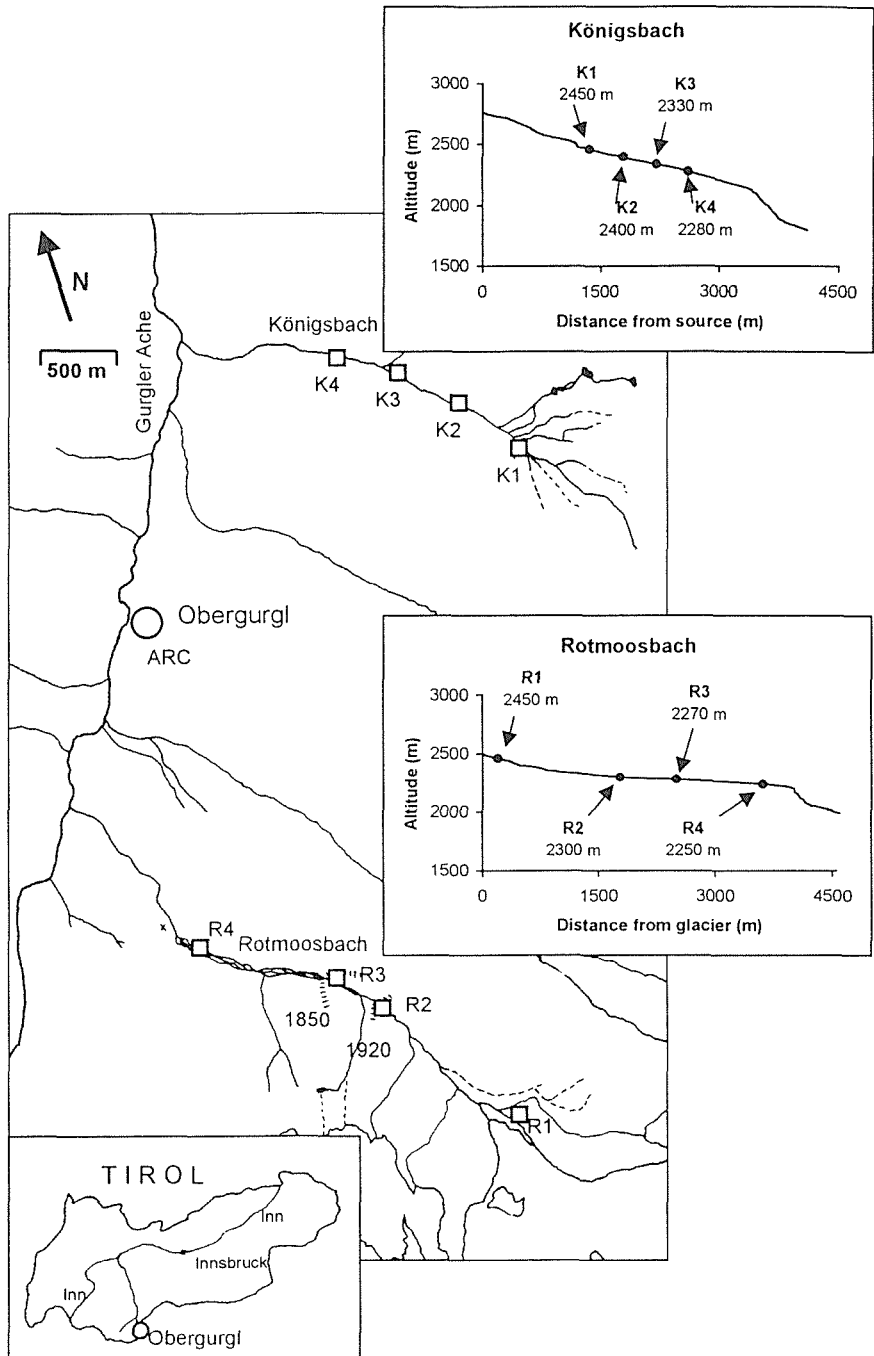


Fig. 1 Map of the investigation area near Obergurgl (Ötztal, Tyrol) and the two study streams.

obtained from field samples (water samples were filtered through pre-washed Whatman GF/C filters) followed by subsequent analysis in the laboratory. Monthly measurements of water chemistry included alkalinity, ammonia, nitrate, sulphate, chloride, total phosphorus, dissolved phosphorus, dissolved organic carbon, sodium, potassium, calcium and magnesium. Replicate samples were in most cases either collected twice each day at maximum and minimum flow or at diurnal measurements every 2 h.

Quantitative samples of benthic invertebrates were obtained by using a "Surber sampler" (five replicates randomly taken at each sampling site and each sampling occasion; mesh size 100 μm). Four emergence traps per sampling site (rectangular base 50 $\times$ 50 cm, pyramid type) have been exposed in summer 1996; except at the glacier mouth. During the snow-free period they were emptied every one or two weeks.

PRELIMINARY RESULTS

Physico-chemical characterization of the high Alpine streams

Compared to spring-fed streams, running waters originating from glaciers show strong diurnal and seasonal fluctuations in discharge, concentrations of suspended solids and ionic composition. Also in the glacier dominated Rotmoosbach, diurnal amplitudes in discharge were highest at strongest glacier ablation in summer (Fig. 2(c)), whereas in winter discharge was low. In Königsbach increases of discharge were associated with snowmelt and rain events, however, streamflow remained relatively constant throughout the year.

Temperature differed within (between the sites) and between the streams respectively (Fig. 2(a)). After snow-break in May temperatures at R2 increased and reached their maximum in July. Site R4 remained uncovered by snow for the whole winter (at very low discharge temperatures below 0°C) hence a small increase of mean temperatures was recorded from January through March. Beginning in March, mean temperatures increased. In Königsbach a faster increase was detectable at all sites, which also continued when at Rotmoosbach temperatures—due to enhanced glacier ablation—decreased.

Maximum discharge due to strong glacier ablation in summer resulted in higher concentrations of suspended solids and also caused increases of pH, as well as particulate and dissolved phosphorus in the glacial stream. Nevertheless, suspended solids were dependent on weather conditions (compare August values in Fig. 2(d)). Maxima in Königsbach occurred at snow-break, but were by three orders of magnitude lower than in the glacial stream. In winter suspended organic matter did not differ within both streams ($< 1 \text{ mg l}^{-1}$), but increased in the glacial stream with increasing discharge and glacier ablation to much higher values (10 mg l^{-1} , maximum in August 45 mg l^{-1} ; Fig. 2(e)). Nitrate ($\text{NO}_3\text{-N}$) values did not differ between streams, however annual means of ammonium ($\text{NH}_4\text{-N}$) were higher in Rotmoosbach by one order of magnitude (Table 1).

Conductivity, generally low in glacial streams, can—depending on the season when the proportion of groundwater inflow is higher—also increase (e.g. in winter

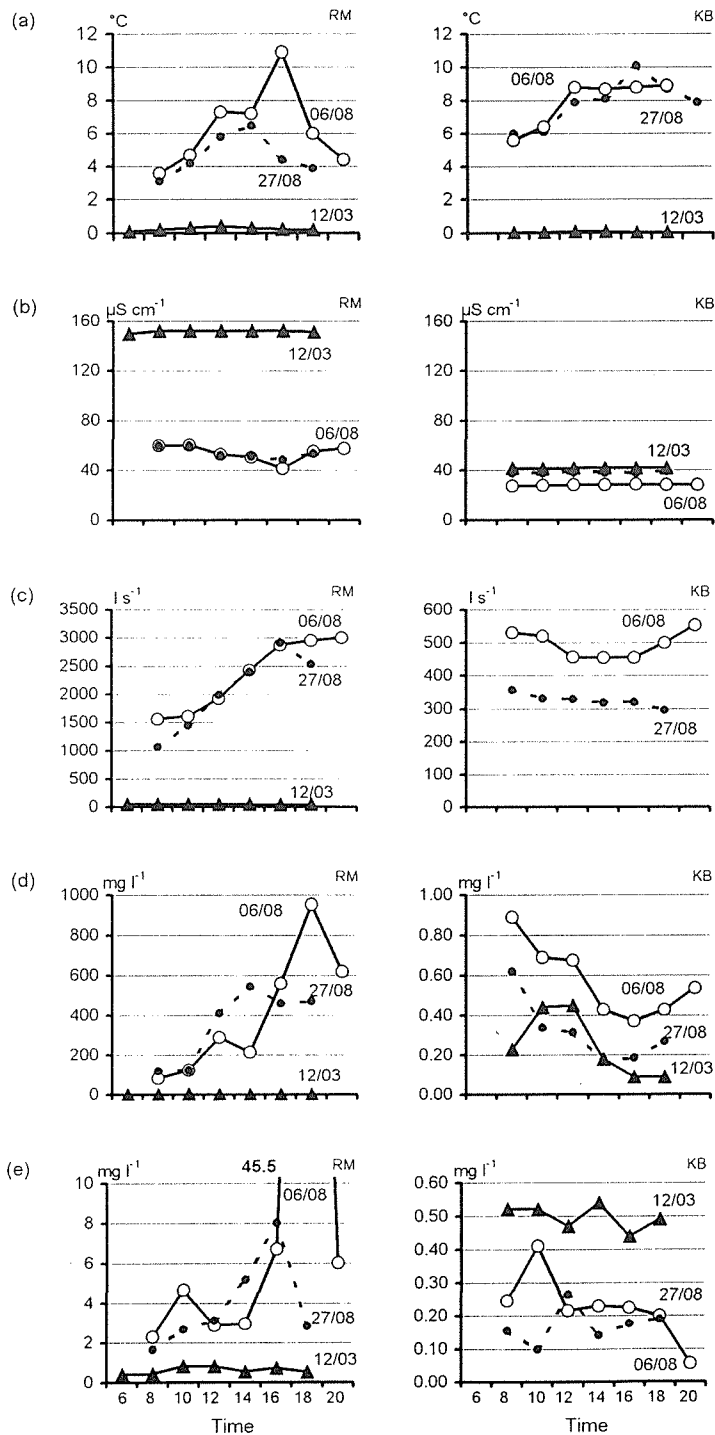


Fig. 2 Diurnal measurements of (a) temperature, (b) conductivity, (c) discharge, (d) suspended solids and (e) organic matter at three sampling dates (RM Rotmoosbach, KB Königsbach).

Table 1 Chemical characterization of Rotmoosbach (R1-R4), its tributaries (A-D), and Königsbach (K1-K4); means are annual means, STD standard deviation; also diurnal fluctuations, expressed as diurnal means and standard deviation.

Parameter	Rotmoosbach:				Tributaries:				Königsbach:				Diurnal fluctuations:			
	R1:	R2:	R3:	R4:	A:	B:	C:	D:	K1:	K2:	K3:	K4:	R4	R4	K4	K4
	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD
Conductivity ($\mu\text{S cm}^{-1}$)	74.6	52.0	147.6	88.8	122.5	75.3	112.4	48.3	117.6	42.9	170.2	40.4	45.1	18.8	48.1	13.5
pH (min-max)	6.83-9.27	7.72-8.98	7.7-8.90	6.6-8.71	7.75-8.17	7.83-8.31	5.74-7.22	6.18-7.08								
Alk Gran ($\mu\text{eq l}^{-1}$)	554.5	253.7	911.6	450.8	764.9	412.1	716.4	292.1	737.3	253.5	1226	487.4	8.2	7.6	67.0	133.8
$\text{NO}_3\text{-N}$ ($\mu\text{g l}^{-1}$)	164.5	97.1	184.5	65.2	178.8	45.0	155.8	39.8	193.4	36.5	42.2	32.1	119.6	30.3	169.4	34.1
$\text{NH}_4\text{-N}$ ($\mu\text{g l}^{-1}$)	35.5	19.9	23.6	17.3	28.1	17.6	18.9	15.7	10.7	5.9	2.1	1.7	34.3	10.8	12.2	12.3
P_r ($\mu\text{g l}^{-1}$)	406.8	380.2	112.4	187.8	172.5	229.6	98.4	150.5	50.0	53.9	16.2	20.2	191.4	210.1	104.1	99.7
P_b ($\mu\text{g l}^{-1}$)	3.0	4.5	1.8	1.7	1.6	1.6	1.6	1.3	1.1	1.1	1.0	0.8	1.4	1.8	0.5	0.7
P_p ($\mu\text{g l}^{-1}$)	403.9	381.5	110.7	188.0	159.6	225.8	96.9	150.6	48.9	54.4	15.2	20.6	190.0	208.7	103.6	99.8
DOC (mg l^{-1})	0.43	0.12	0.37	0.11	0.40	0.14	0.59	0.34	0.41	0.17	0.47	0.18	0.39	0.18	0.32	0.13
SO_4^{2-} (mg l^{-1})	8.75	9.45	20.23	16.76	16.18	13.05	13.99	6.77	12.50	6.22	15.74	2.74	13.40	5.28	12.26	5.92
Cl^- (mg l^{-1})	0.46	0.80	0.30	0.32	0.26	0.18	0.97	1.97	0.27	0.33	0.15	0.12	0.36	0.54	1.07	1.72
Na^+ (mg l^{-1})	0.33	0.21	0.62	0.43	0.48	0.31	0.49	0.22	0.25	0.13	0.60	0.08	0.49	0.36	0.45	0.20
K^+ (mg l^{-1})	1.02	0.96	2.31	1.67	1.82	1.34	1.78	0.89	1.45	0.79	2.48	0.72	1.04	0.39	1.08	0.43
Mg^{2+} (mg l^{-1})	1.89	2.18	4.10	3.21	2.99	2.46	2.62	1.43	3.03	2.13	5.16	1.98	1.40	0.60	1.05	0.50
Ca^{2+} (mm l^{-1})	10.67	5.46	19.66	11.24	16.82	9.77	16.09	6.38	15.87	4.12	24.31	7.15	4.00	1.63	4.36	1.28
Königsbach:																
Diurnal fluctuations:																
K1:																
Conductivity ($\mu\text{S cm}^{-1}$)	40.0	11.9	32.8	10.4	33.3	9.1	33.0	8.2	153.6	0.3	53.9	6.6	41.7	0.1	28.1	0.5
pH (min-max)	6.48-6.93	6.49-7.79	6.52-7.27	6.54-7.44	7.77-7.82	7.65-8.30	6.54-7.44	7.65-8.30								
Alk Gran ($\mu\text{eq l}^{-1}$)	48.7	10.1	47.1	6.8	50.4	8.2	50.9	8.6	70.7	6.9	97.4	98.0	1038.6	3.8	563.7	52.2
$\text{NO}_3\text{-N}$ ($\mu\text{g l}^{-1}$)	202	89	174.6	38.7	185.5	54.6	176.8	69.4	133.6	31.7	108.7	18.7	170.0	0.0	97.0	5.4
$\text{NH}_4\text{-N}$ ($\mu\text{g l}^{-1}$)	2.5	2.2	2.2	1.6	1.8	1.6	6.5	13.2	1.5	0.8	32.1	13.0	3.8	5.5	3.1	3.5
P_r ($\mu\text{g l}^{-1}$)	5.3	6.1	2.7	2.1	2.4	1.8	2.0	1.6	3.2	1.7	301.5	206.5	1.5	0.2	15.9	18.3
P_b ($\mu\text{g l}^{-1}$)	0.9	0	1.1	0.9	1.1	1.0	0.9	0.6	1.4	0.2	12.5	7.9	1.2	0.1	2.5	1.1
P_p ($\mu\text{g l}^{-1}$)	4.36	5.7	1.7	2.0	1.2	1.3	1.2	1.5	1.8	1.7	289.0	199.3	0.3	0.2	13.4	18.1
DOC (mg l^{-1})	0.30	0.08	0.30	0.07	0.28	0.05	0.29	0.05	1.10	0.41	0.45	0.08	0.40	0.07	0.37	0.09
SO_4^{2-} (mg l^{-1})	9.91	2.79	9.41	3.51	9.48	3.25	8.89	2.24	23.33	10.47	6.25	1.10	10.10	0.11	7.51	0.14
Cl^- (mg l^{-1})	0.17	0.09	0.20	0.26	0.14	0.05	0.18	0.10	0.23	0.04	0.48	0.48	0.43	0.57	0.25	0.39
Na^+ (mg l^{-1})	0.38	0.08	0.34	0.09	0.39	0.09	0.43	0.10	0.76	0.02	0.44	0.18	0.86	0.02	0.38	0.02
K^+ (mg l^{-1})	0.71	0.12	0.66	0.13	0.69	0.13	0.69	0.11	2.57	0.04	0.73	0.12	0.77	0.05	0.64	0.02
Mg^{2+} (mg l^{-1})	0.91	0.33	0.74	0.28	0.74	0.24	0.72	0.23	3.70	0.04	1.09	0.17	0.89	0.02	0.58	0.01
Ca^{2+} (mg l^{-1})	4.22	1.45	3.64	1.27	3.65	1.16	3.60	1.07	22.91	0.09	9.07	0.58	4.67	0.04	2.92	0.29

$153.6 \pm 0.3 \mu\text{S cm}^{-1}$; see also Fig. 2(b)). According to the catchment geology consisting of calcareous rocks conductivity in Rotmoos was higher than in glacial streams in general (Table 1). Even at the glacial snout pronounced fluctuations occurred ($74.6 \pm 52.0 \mu\text{S cm}^{-1}$).

Dissolved organic carbon and phosphate values also demonstrate the interplay of groundwater and glacier ablation. Whereas Rotmoosbach showed seasonal differences (diurnal mean in March $1.10 \pm 0.41 \text{ mg l}^{-1}$ in August $0.45 \pm 0.08 \text{ mg l}^{-1}$ DOC), in Königsbach DOC was low throughout the year (in March 0.40 ± 0.07 compared to August $0.37 \pm 0.09 \text{ mg l}^{-1}$).

Total phosphorus values (mainly particulate phosphorus) were two orders of magnitude higher than in Königsbach (at the glacier snout $406.8 \pm 380.2 \mu\text{g l}^{-1}$; concentrations decreasing with increasing distance from the glacier margin). High phosphate concentrations are characteristic for glacial flour and are known from other investigations in glacial streams (Bretschko, 1966). Also in Rotmoosbach, increasing phosphate concentrations were always tied to increasing discharge.

Differences in pH values were distinct, too. Here as well, especially at times of maximum glacier ablation, pH values increased (> 8) in the glacial stream (Table 1). Low values were measured in both streams, especially at rainy periods or at snow-break.

Insects and invertebrates in high altitude streams

The fauna of the compared streams differs in many respects but also shows some common aspects. In both streams faunal elements, typical for high Alpine streams, were found. The number of taxa in the glacial stream was lower than in the spring-fed stream (29 compared to 56 so far). The abundance of aquatic insect larvae was different in both streams, in longitudinal as well as in temporal respects. Rotmoosbach showed lower densities in all samples analysed so far. The spring-fed stream inhabits higher densities of aquatic insects, dominated by the dipteran larvae of Orthoclaadiinae and Diamesinae. However, Ephemeroptera, Plecoptera and Trichoptera already appear in the highest reach of the stream (for more details see Schütz *et al.*, 1998).

Similar findings are recorded from other studies in high mountain streams. The dominance of the larvae of the genus *Diamesa* is a common feature of high mountain streams above the upper forest boundary (e.g. Bretschko, 1969; Kownacki, 1991; Ward 1992, 1994). With increasing distance from the glacier interference with the terrestrial environment becomes more and more important (runoff from Alpine grasslands; influence of groundwater, springs, tributaries, etc.). Accordingly, a change in community structure and trophic relationships of colonizing invertebrates occurs. Our findings generally follow these models and statements.

The densities of emerging aquatic insects differed also between the two streams. Especially in summer the mean emergence rate (all taxa) was about 10 times higher in Königsbach. In the glacial stream it remained generally low (usually $< 10 \text{ Ind. m}^{-2} \text{ day}^{-1}$). Only at our lowest site (R4) similar densities were found. At all sites a distinct increase occurred in August. In the glacial stream minor peaks

occurred already in May and June, whereas at this time Königsbach was still covered with snow (unpublished data).

CONCLUSIONS

Our results describe high altitude stream types in terms of the prevailing fauna, species composition, taxa number, abundance and relevant abiotic conditions. The results presented here are tentative, despite the fact that temporal and spatial differences in physico-chemical conditions as well as biological implications were distinct. In a final stage of our still ongoing investigation we would like to answer open ecological questions, such as the adaptation of the fauna to specific abiotic features. Field experiments planned for 1998 will approach possible effects of substrate stability, organic matter, biofilm development, and nutrient quality on the composition of stream benthos.

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Chemistry of headwater lakes studied in the EU project “Acidification of mountain lakes: palaeolimnology and ecology (AL:PE)”

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Abstract The AL:PE project, involving research institutes from 11 countries, was launched in 1991 to study sediments, diatoms, fish, invertebrates, and water chemistry of 28 lakes in relation to atmospheric pollutant inputs. This paper discusses the chemical results and the analytical quality control used to ensure good comparability of the results. Lake waters show varied chemical characteristics—from very diluted, poorly buffered or acidic waters to lakes with high base cation concentrations—with electric conductivity values ranging from 0.4 to 6 mS m⁻¹ at 25°C. Nitrate shows a high range of variability, from values close to zero (Svalbard and Portugal), to 11–25 µeq l⁻¹ (Alps), and up to 50 µeq l⁻¹ (Tatra Mountains).

INTRODUCTION

High mountain lakes are sensitive to any input of pollutants due to climatic factors, shallow soil cover, small basins, and rapid flush rates. They can thus be regarded as “early warning” indicators for any variation in the input of atmospheric pollutants as well as for climatic changes. On this basis, an interdisciplinary research project examining 28 lakes throughout most of the European remote areas (Table 1 and Fig. 1) was started in 1991 with the participation of 14 institutes from 11 countries (Table 2).

The project was aimed towards an understanding of remote mountain lake ecosystems and their response to varying levels of deposition of acid compounds and other air pollutants (e.g. trace metals, PCBs) transported over long ranges, as well as at identifying unpolluted remote lakes within Europe to be consequently used as reference sites for climate change studies.

The research activities were organized in different subject areas, concerning water chemistry (main ions, nutrients, trace metals), biology (epilithic diatoms, zooplankton, macroinvertebrates and fish), sediment (dating, trace metals, persistent trace organics, carbonaceous particles, diatoms, chironomids) and statistical analysis and modelling.

Table 1 Altitude, pH, conductivity at 25°C (Cond, $\mu\text{S cm}^{-1}$) and ion concentrations ($\mu\text{eq l}^{-1}$) of the AL:PE 2 lakes, grouped according to results from a cluster analysis of chemical properties.

Group	Code	Lake	Altitude	pH	Cond	NH ₄ ⁺	Ca ⁺⁺	Mg ⁺⁺	Na ⁺	Alk	SO ₄ ⁼	NO ₃ ⁻	Cl ⁻
1	19	Zgornje Krisko J.	2150	7.52	5.7	5	368	66	31	371	138	7	6
2	9	Arresjøen	20	5.81	3.8	0	35	49	207	24	32	0	233
2	10	Maam	436	4.96	6.0	0	40	63	341	7	61	3	392
3	18	Chibini	474	7.47	3.4	0	40	8	196	170	55	10	23
4	7.4	Noir	2750	6.96	1.9	0	125	31	13	125	46	9	6
4	13	La Caldera	3050	7.34	3.3	7	278	44	12	346	37	1	62
4	15.2	Nizne Terianske Pl.	1941	6.40	1.9	0	143	7	16	68	55	31	5
4	15.4	Zieloni Staw	1671	6.57	2.2	2	146	15	19	75	72	34	7
5	3	Lille Hovvatn	503	4.69	2.5	0	21	13	53	9	54	11	53
5	4.1	Lochnagar	785	5.25	2.3	3	39	43	101	-4	57	22	96
5	4.2	Sandy Loch	790	5.05	1.8	0	52	34	88	9	53	8	65
5	4.3	Nan Eun	890	5.06	1.9	0	34	25	76	-7	49	19	73
6	5.2	Paione Inferiore	2002	6.53	1.3	1	73	12	14	25	48	25	4
6	6.1	Lungo	2384	6.17	1.4	2	69	13	12	27	60	16	3
6	6.2	di Latte	2540	6.35	1.6	2	92	11	13	51	56	22	3
6	7.3	Blanc	2753	6.91	1.3	2	75	24	29	83	31	11	55
6	11	Schwarzsee ob Sölden	2799	5.60	1.6	2	68	24	16	7	86	11	3
6	12.2	Estany Redo	2240	6.69	1.2	1	70	12	10	44	23	11	8
6	15.3	Dlugi Staw	1783	5.95	2.0	1	116	10	17	18	75	50	7
7	1	Øvre Neådalsvatn	728	6.17	1.1	0	28	11	40	32	18	1	34
7	2	Stavsvatn	1053	6.00	0.7	1	40	9	21	16	33	3	17
7	5.1	Paione Superiore	2269	5.85	1.0	3	42	7	8	1	36	21	4
7	8	Etang d'Aubé	2091	6.11	0.7	2	28	6	16	30	17	16	9
7	12.1	Estany Aguilò	2210	6.10	0.5	0	18	10	9	11	13	2	7
7	14	Escura	1680	5.81	1.0	2	19	12	43	5	17	0	52
7	15.1	Starolesnianske Pl.	2000	4.69	1.4	3	29	6	3	0	52	10	3
7	16	Laguna Cimera	2140	5.69	0.4	0	14	9	30	17	11	20	9
7	17	Chuna	420	6.45	1.6	2	64	17	37	41	50	4	28

METHODS

Lake water was sampled at monthly or annual intervals. Samples were taken either at the outlet of the lake or at the surface in the deepest part of the lake and analysed by the participating laboratories with different methods harmonized through yearly intercomparison exercises (Wathne *et al.*, 1995, 1997). Cluster analysis was performed using Ward's method and 1-Pearson's *r* as the distance measure.

RESULTS

The yearly mean pH values range from 4.7 in the most acidic lakes to values above 7.0

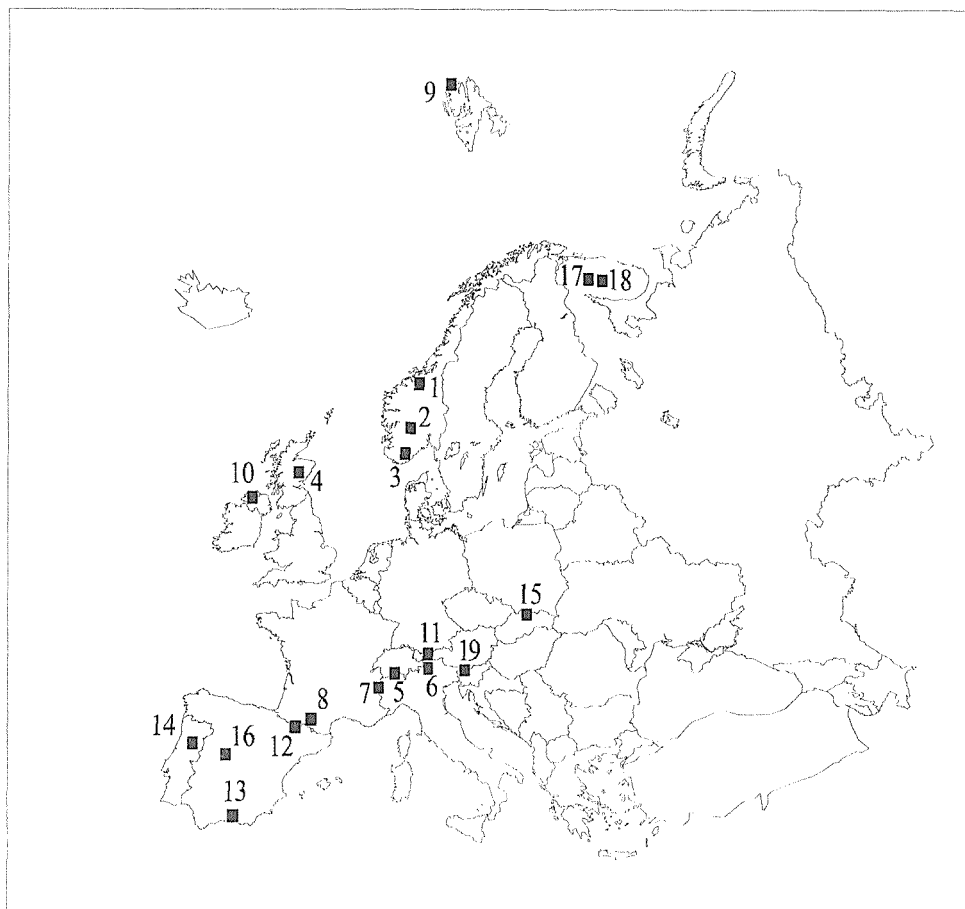


Fig. 1 Location of the AL:PE 2 lakes. Numbers refer to Table 1.

in Chibini, La Caldera, and Zgornje Krisko (Table 1), with an evident gradient existing in Norway from the unpolluted Øvre Neådalsvatn with pH value of 6.2 to the medium exposed Stavsvatn (pH 6.0) to Lille Hovvatn (pH 4.7), the site most heavily exposed to acid rain. Electric conductivity values range from 0.4 to 6.0 mS m^{-1} at 25°C; the highest value (Maam) is due to a high marine contribution of solutes.

Figure 2 shows the SO_4^- and NO_3^- concentrations for the AL:PE lakes. Again, the pollution gradient in Norway from Øvre Neådalsvatn to Lille Hovvatn is clearly visible. The SO_4^- concentrations range from 30 to 60 $\mu\text{eq l}^{-1}$ in 17 lakes (61%). The highest value of 138 $\mu\text{eq l}^{-1}$ in Zgornje Krisko (Slovenia) is due to the geology of the basin. The lowest SO_4^- values, below 20 $\mu\text{eq l}^{-1}$, are found in the French Pyrenees, in the Iberian peninsula, and in the reference remote site Øvre Neådalsvatn. Nitrate concentrations in the Tatra are the highest. Further gradients might exist from the north (Svalbard and Norway) to central Europe (via UK to Italy and the Tatra) and from west to east, with low values in Portugal, Spain, and the Alps to high values in the Tatra Mountains.

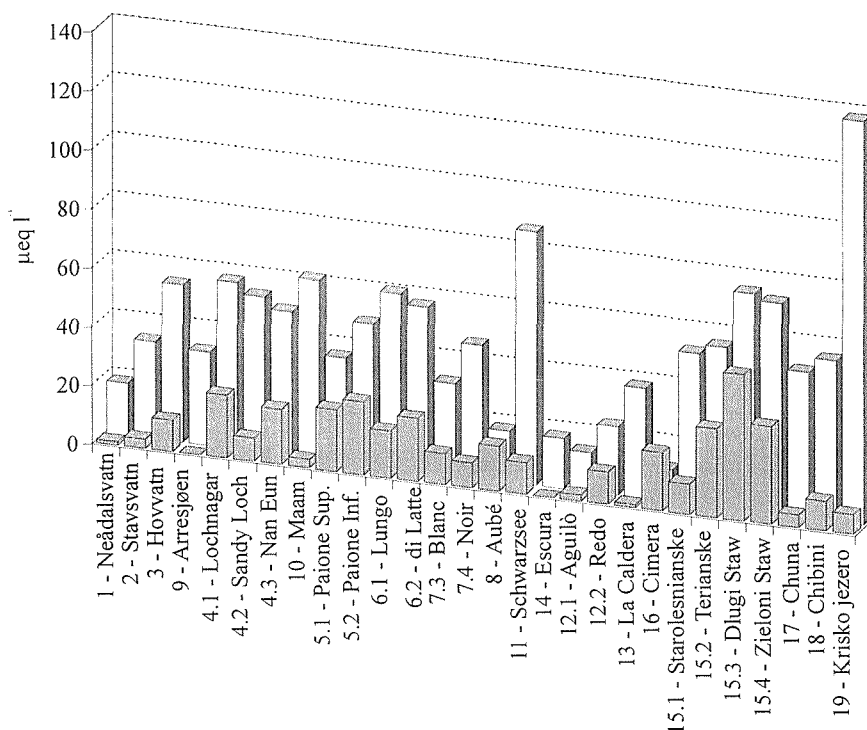
The lakes were grouped on the basis of their ion concentrations using cluster

Table 2 Institutions participating in the AL:PE 2 project and respective lakes. Numbers refer to Table 1.

Institute involved	Lake codes
Norwegian Institute for Water Research, Oslo (N)	1, 2, 3, 9
University College London, Environ. Change Research Centre (UK)	4.1, 4.2, 4.3, 10
Istituto Italiano di Idrobiologia, Pallanza (I)	5.1, 5.2
Laboratorio Biologico Provinciale, Laives (I)	6.1, 6.2
Univ. Bordeaux, Lab. Neurobiol. Physiologie Comparées, Arcachon (F)	7.3, 7.4, 8
University of Innsbruck, Inst. of Zoology and Limnology (A)	11
Universidad de Barcelona, Dept Ecologia (E)	12.1, 12.2, 14*, 16
Universidad de Granada, Instituto del Agua (E)	13
Charles University, Dept Hydrobiology, Prague (CZ)	15.1, 15.2
Inst. Zoology, Slovak Academy Sciences, Dept Hydrobiology, Bratislava (SK)	15.1, 15.2
Inst. Freshwater Biology, Polish Academy of Sciences, Cracow (PL)	15.3, 15.4
Universidad Autonoma Madrid, Dept Ecologia (E)	16
Inst. Ecological Problems, Kola Science Centre, Apatity (SU)	17, 18
National Institute of Biology, Ljubljana (SLO)	19

* Site 14 is located in Portugal.

analysis in seven different categories (Table 1). Two of them (groups 1 and 3) are composed of single lakes with peculiar lithology of the basin: Zgornje Krisko, draining a calcareous basin, and Chibini with alkaline rocks (apatite, nepheline syenites) in the basin (Moiseenko, 1994). Group 2 is characterized by a high sea salt contribution, with

**Fig. 2** Average concentrations of sulphate (white bars) and nitrate (dark bars) for AL:PE 2 lakes (1993–1995).

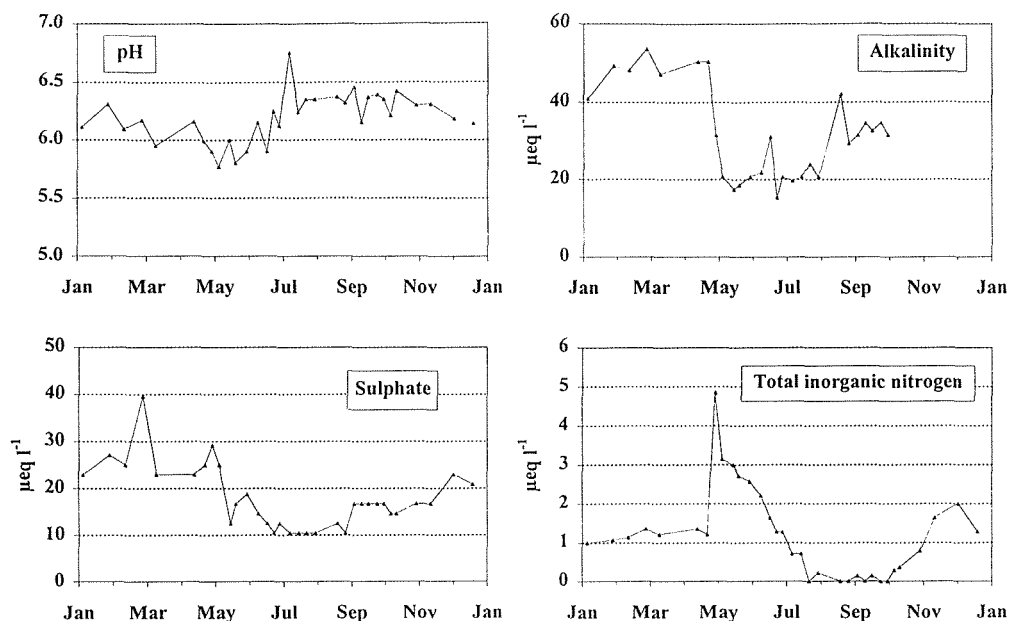


Fig. 3 Seasonal variation of pH, alkalinity, sulphate, and total inorganic nitrogen in Øvre Neådalsvatn for the year 1994.

a mean Na/Cl ratio of 0.88, very close to mean seawater value (0.86). Group 5 shows relatively high sodium and chloride concentrations, and higher Na/Cl mean ratio (1.04), indicating a contribution of sodium from silicate weathering. Group 4 is formed by lakes with high alkalinity, although lower than in groups 1 and 3. Groups 6 and 7 contain lakes characterized by low ionic content, with the prevalence of calcium, alkalinity and $\text{SO}_4^{=}$, and higher $\text{SO}_4^{=}$ and NO_3^- levels in the first.

Seasonal variation of solute concentrations in headwater lakes is an important topic also in relation with the effects on biota. Snowmelt is the most significant period, for the high flux of water and pollutants to the lake, which are responsible for pH and alkalinity minima. This is evident in May in Neådalsvatn (Fig. 3), where alkalinity and pH drop from 50 to 18 $\mu\text{eq l}^{-1}$ and from 6.2 to 5.7, respectively and $\text{SO}_4^{=}$ and NO_3^- concentrations increase. During summer, alkalinity increases as an effect of weathering and lower water fluxes, while inorganic nitrogen, used by phytoplankton, reaches the lowest levels. A similar pattern is shown by Paione Superiore (Fig. 4) with higher concentrations of inorganic nitrogen (16–32 $\mu\text{eq l}^{-1}$).

Although not included in the AL:PE project, the trend of chemical concentrations can be analysed by pooling several lakes. In the case of Stavsvatn (Fig. 4), pH and alkalinity increase during the period 1986–1997, while $\text{SO}_4^{=}$ concentration decreases and inorganic nitrogen shows strong fluctuations, with a slight increase during the last few years. Although less evident, the same trends are shown in Paione Superiore in the period between 1984 and 1996. In both cases, these trends are related to a decrease of acidity and $\text{SO}_4^{=}$ content of atmospheric deposition, whereas nitrogen is constant or in slight increase (Mosello *et al.*, 1993; Henriksen *et al.*, 1990).

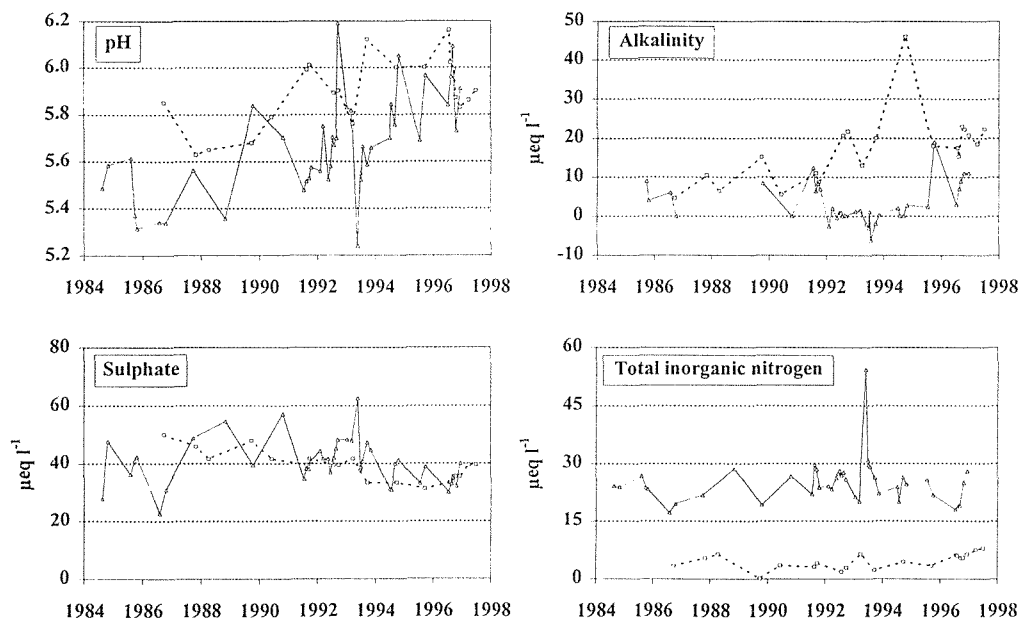


Fig. 4 Trend of pH, alkalinity, sulphate, and total inorganic nitrogen in Paione Superiore (full line) and Stavsvatn (broken line), for the period 1984–1996 and 1986–1997, respectively.

CONCLUSIONS AND PERSPECTIVES

The most interesting aspect of the AL:PE project was the great attention paid to the methodological aspects with the formalization in a manual (Wathne, 1996) of common plans to assure the quality of data, regular intercomparison exercises, and exchanges between experts with joint sampling and laboratory activities.

The AL:PE lakes represent a wide range of remote European areas with different levels of pollution, as long range pollution in Europe reaches most of the remote sites. The results showed interesting similarities among the sites at all levels indicating that the biological communities in these high-altitude and latitude areas react in the same way to a specific chemical environment.

The discussion of the data collected has provided indications of the topics and the areas which require to be studied in greater detail. All these aspects have been formalized in a research project “*Measuring and modelling the dynamic response of remote mountain lake ecosystem to environmental change: a programme of mountain lake research* (MOLAR)”, at present ongoing, which saw an increase in the number of participating laboratories (from 14 to 19) and countries (from 11 to 14).

This programme includes (a) the evaluation of critical and present loads of sulphur, nitrogen, and other pollutants, such as PAHs and PCBs and of the deposition of dust in high altitude areas, (b) the identifying of biological variables and processes which can serve as early warning indicators and (c) the assessment of the importance of the microbial loop in the pelagic food web in lakes with different degrees of acidification.

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Measuring and modelling the dynamic response of remote mountain lake ecosystems to environmental change: an introduction to the MOLAR project

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Abstract Because of their sensitivity, remote mountain lakes are not only vulnerable to environmental change, they are also excellent sensors of change, and their high quality sediment records can be used to infer the speed, direction, and biological impact of changing air quality and climate. The MOLAR project focuses on detailed studies of a smaller number of key sites to provide high resolution data on their temporal dynamics that can then be used to develop and calibrate predictive models.

INTRODUCTION

The research project MOLAR is funded within the EU Environment and Climate Programme with assistance from INCO. It is coordinated by the Environmental Change Research Centre (ECRC) at University College London and the Norwegian Institute for Water Research (NIVA) and involves the collaborative participation of 23 laboratories throughout Europe. Although it is too early (the project was initiated in February 1996) to present results, this paper presents the rationale and objectives of MOLAR.

The Arctic and Alpine regions of Europe represent the most remote and least disturbed environments in Europe, yet they are threatened by acid deposition, toxic air pollutants, and by climate change. The remote lakes that occur throughout these regions are especially sensitive to these threats for a number of related reasons:

- many have little ability to neutralize acidity because of their low base status;
- nitrate levels are higher because their catchments have little soil and vegetation to take up nitrogen deposition;
- toxic trace metals and trace organics accumulate in the food chain more easily, and some pollutants (e.g. mercury, volatile organics) accumulate preferentially in cold regions;

- climatic warming in Europe is predicted to be greatest in Arctic and Alpine regions.

Because of this sensitivity, remote mountain lakes are not only vulnerable to environmental change, they are also excellent sensors of change, and their high quality sediment records can be used to infer the speed, direction, and biological impact of changing air quality and climate.

The MOLAR project builds on the EU funded projects AL:PE.1 (Acidification of Remote Mountain Lakes: Palaeolimnology and Ecology) and AL:PE.2 (Remote Mountain Lakes as Indicators of Air Pollution and Climate Change), which represented the first comprehensive study of remote mountain lakes at a European level and assessed the status of these ecosystems on the basis of sediment core records and chemical and biological surveys. (Wathne *et al.*, 1995, 1997; Battarbee *et al.*, 1997).

OBJECTIVES

The project has four overall objectives, each corresponding to a major strand or “work package” in the work programme:

- (1) to measure and model the dynamic responses of remote mountain lake ecosystems to acid (sulphur plus nitrogen) deposition;
- (2) to quantify and model pollutant (trace metals, trace organics) fluxes and pathways in remote mountain lakes and their uptake by fish;
- (3) to measure and model the temporal responses of remote mountain lake ecosystems to climate variability on seasonal, inter-annual and decadal time scales;
- (4) to continue the development of a high quality environmental database on remote mountain lake ecosystems in Europe and to disseminate results widely.

The main output generated by this project will be the development of predictive models for acidity, pollutant flux, and climate variability that can be used in scenario assessment studies, especially those scenarios associated with present and forthcoming UNECE protocols and General Circulation Model (GCM) predictions for Europe.

STUDY SITES

All sites strictly adhere to the characteristics established for AL:PE, namely that they should be above or beyond the regional timberline and have no evidence of human disturbance in the lake or catchment area. Thus, any changes in their ecology can only be due to air pollution, climate change, and natural variability.

MOLAR concentrates on a selected number of key sites (Fig. 1), but in contrast to AL:PE where sites were sampled only once per year, MOLAR sites are being sampled intensively throughout a period of 18–24 months to provide data for modelling. Data from less-intensively sampled secondary sites (Fig. 1) and from other AL:PE sites not included for specific study within MOLAR will be used to enhance model validation.

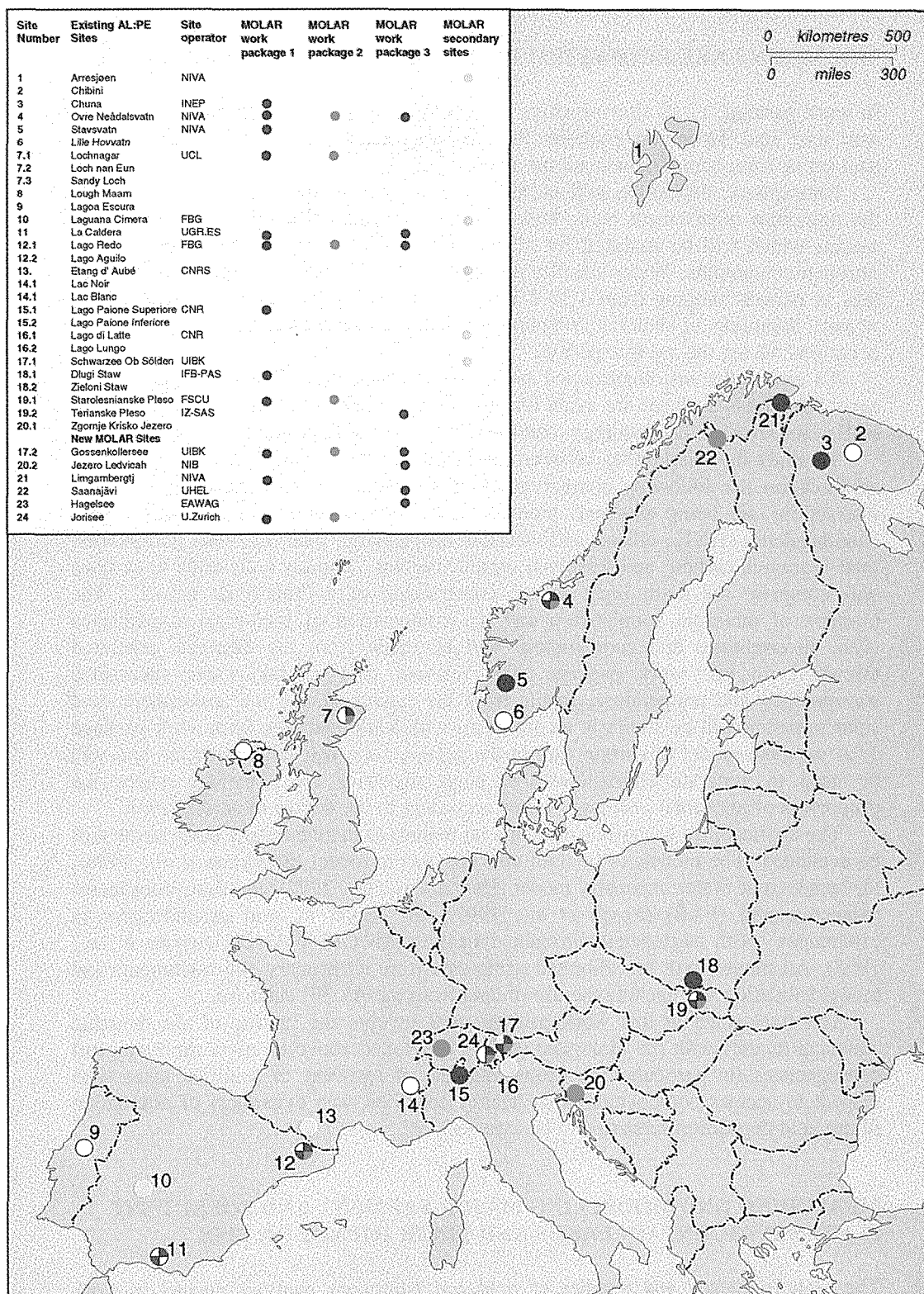


Fig. 1 MOLAR sites: distribution, location of work packages and relationship with the AL:PE network.

MOUNTAIN LAKE ECOSYSTEM RESPONSE TO ACID DEPOSITION

In work package 1, 12 sites sensitive to acidification located along north–south and west–east acid deposition gradients in Europe have been selected from the AL:PE data set or from comparative national studies (Fig. 1).

Atmospheric deposition will be measured at each of the 12 primary lakes with the minimum requirement being bulk-collected samples collected at periods of 2–4 weeks, which will be analysed for all major ions with particular attention given to nitrogen compounds. Water samples are taken from the surface at the outlet of each lake at periods ranging from 1 to 2 weeks in the ice-melt season to bi-monthly in winter. Sampling of diatoms, zooplankton and invertebrates is undertaken on 2–3 occasions during the ice-free period.

To assess the histological and physiological attributes of fish as indicators of early acid stress analyses are made for physiology, histology of gills, surface tissues, livers, and ovaries, haematology, and metallothionein in liver and kidney.

To study the relationship between microbial activity in the pelagic food web and acidification the following compartments of the pelagic system in lakes and their interactions are being assessed: biomass of bacteria, phytoplankton, heterotrophic nanoflagellates (HNF), ciliates and metazoic zooplankton, total dissolved phosphorus, total inorganic carbon, and dissolved organic carbon. Biomass (converted to carbon) will comprise the main input in the construction of a “microbial model”. The biomass of metazoic zooplankton and the composition of biological assemblages (fish, invertebrates and zooplankton) will be taken from the database generated elsewhere in this work package. The relevant data on lake-water chemistry, transparency and temperature, inputs from the catchment and the atmosphere, and solar radiation will be similarly incorporated with additional data from work package 3. A seasonal model of carbon flux in the pelagic zone will be constructed and will be used to generate alternative flow chart scenarios for situations where the proportion of microbial processes varies according to the degree of acidification.

The applicability of various critical load models to mountain lake ecosystems will be evaluated. The models to be used are: the diatom model (Battarbee *et al.*, 1996), the steady-state water chemistry model (Henriksen *et al.*, 1992), the first order mass-balance model (FAB) (Posch *et al.*, 1997), the model of acid groundwaters in catchments—with aggregated nitrogen dynamics (MAGIC-WAND) (Ferrier *et al.*, 1995). All models will be calibrated using data from all primary site catchments and model validation will include the use of data from the AL:PE database.

The final stage in this work package will involve the linking of the dynamic chemical models with the biological models developed above to allow the biological consequences (in particular the extent and rate of recovery of acidified lakes with respect to species composition, fish health, and food web dynamics) of alternative future acid deposition scenarios to be assessed.

MEASURING AND MODELLING MAJOR ELEMENT AND POLLUTANT FLUXES IN MOUNTAIN LAKES AND THEIR IMPACT ON FISH

The high frequency and number of field and laboratory analyses needed in work

package 2 require a sub-set of MOLAR sites which combine remoteness with reasonable ease of access and certain on-site basic facilities (Fig. 1).

Pollutant deposition will be sampled and analysed and the pollutants allocated to source. Bulk deposition of nitrogen and sulphur compounds is carried out in work package 1. Automatic weather stations are installed at all sites to measure temperature, humidity, radiation, precipitation, wind speed and direction. Dry and wet deposition is automatically sampled at daily intervals at the stations where electricity is available. Bulk deposition is sampled at all sites at weekly intervals. Snow and rain are collected monthly and analysed for nutrients, SCPs, PAH, and organochlorinated compounds. Soil cores will be taken from undisturbed sites in the lake catchments to determine the total inventory of ^{210}Pb and ^{137}Cs . Event sampling and back trajectory calculation for the air masses during specified time periods is available at the Austrian, Spanish, and Norwegian stations from the work of national programmes.

To track pollutant pathways within the lake-catchment system lake water is filtered for microscopic, mineralogical, and elemental investigation of particles. For speciation of dissolved substances, large volumes are filtered through resins. Chemical speciation and biogeochemistry of copper, zinc, iron, lead, and cobalt is carried out on samples collected from various depths in the water column of each lake during early spring, summer, autumn and winter. Sediment particles are collected at all sites at monthly intervals by means of an array of four sediment traps at two different depths. At those non-AL:PE lakes where the information is missing, sediments will be sampled and analysed for metals, organochlorinated compounds, PAH, SCPs, and radiotracers. On the basis of these analyses pollutant budgets will be established, including losses via the outflow and to the bottom sediments.

To investigate the effects of trace metals and organic pollutants on fish, analyses of mercury, cadmium, and lead will be carried out on a large range of fish and tissue types. In addition, the activity of the mixed function oxidase (MFO) and bile analysis is to be measured in fish livers along a European north-south transect. The fish reproductive capacity, histological analyses of gills, liver, and kidney and morphological and physiological parameters of gills and blood will be correlated with the concentration of trace metals in the tissues. In order to assess fish uptake of organic pollutants (PAH, PCB, DDTs, and hexachlorobenzene) the livers from fish sampled at the same time and from the same sites will be analysed and the results correlated with the ecotoxicological results from trace metal analysis.

Pollutant fluxes in lake-catchment systems will be modelled using radiotracers and SCPs. Existing models of pollutant deposition and transport through catchment-lake systems will be reviewed and developed to accommodate the particular features of high mountain lakes. Attention will be paid to the significance of transport from the snow pack during the annual thaw and transport through the water column will take into account factors such as water residence time, chemical speciation, particle size distribution, and concentration. Model outputs will include accumulation in fish and the sediment record. Rainwater and snow pack samples are being analysed for ^{210}Pb activity and soil cores from undisturbed sites will be sectioned and analysed for ^{210}Pb and ^{137}Cs and the inventories used to determine the mean annual ^{210}Pb flux from the historical and contemporary atmospheric ^{137}Cs fallout record. Data from ^{210}Pb dating will be used to quantify outputs to the sediment record. SCPs will be

measured in deposition gauges, sediment traps, and sediment cores. The model will finally be tested using data for other sites from the AL:PE/MOLAR project database.

CLIMATE VARIABILITY AND ECOSYSTEM DYNAMICS AT REMOTE MOUNTAIN LAKES

Ten regions of Europe have been selected for new harmonization and correlation studies of weather records over the last 200 years. In each region a local network of stations has been identified which will allow the construction of an independent climate series. Following harmonization, time series and spatial analysis of the data sets will be used to provide a base for validation work across the full range of remote sites. Correlation between on-site weather data and data from nearby lowland meteorological stations is being made to enable short and long-term instrumental data records from lowland sites to be used to interpret data at remote MOLAR sites. Monthly, weekly, and some daily series of temperature, rainfall, and river discharge will be analysed from Europe's mountain ranges. The relationship between site-specific weather patterns and those recorded at national meteorological stations is being assessed using data from automatic weather stations positioned at each site. Precipitation gauges have been set up to measure rainfall and snowfall. In addition, essential ground-truth work to relate data on the freezing and thawing of ice of the lakes to local and regional air temperature data is being carried out to determine historical freeze-up and break-up dates of the MOLAR lakes and to evaluate the usefulness of the freeze-up and break-up dates of high-altitude and high-latitude lakes as proxy air temperature data.

To assess how lake chemistry and biological communities respond to changes in ice-cover, light penetration, temperature change, stratification, and mixing, water column data on temperature, oxygen, pH, conductivity, and chlorophyll-*a* are collected. Analyses for major ion and nutrient chemistry, phytoplankton, and zooplankton are made, and sediment traps are deployed in order to assess the quantity and timing of material fluxes to the sediment, both of autochthonous and allochthonous material.

A series of workshops has been organized to harmonize taxonomic conventions for diatoms, chrysophytes, cladocera, and chironomids and to develop statistical models relating the distribution of key taxa to their environmental (temperature, pH, nutrients) optima and tolerances in mountain lakes.

To establish long-term variability in ecosystem dynamics from recent palaeolimnological records core samples of the upper sediment have been taken and analysed for lithostratigraphy, mineralogy, and magnetic susceptibility to characterize changes in catchment erosion. The sediments are being dated using ^{210}Pb , supplemented by ^{137}Cs and ^{241}Am . Sub-samples from each core section are utilized for pigments and C:N ratio analyses. Training-set data will be compiled in a relational database and transfer functions will be developed for lake-water pH, nutrients, and temperature using weighted averaging regression and calibration. Statistical and ecological evaluation of all reconstructions will be made using modern analogue matching, pooriness-of-fit measures, and root mean squared errors. Finally, climate reconstructions based on the sediment core records will be validated by correlation with long-term instrumental records.

A dynamic model of mountain lake ecosystems will be developed relating weather forcing with water column chemistry and biology and the sediment record. The model will then be used to hindcast responses to climate change in the recent past. These predictions will be validated using the 200 year instrumental weather record (as model inputs) and the sediment record as a proxy reference for output. The model will then be used to forecast responses to alternative future climate scenarios. A full sensitivity analysis of the model will also be carried out.

INTEGRATING ACTIVITIES

Work package 4 is concerned with the standardization and quality control of field and laboratory methods; database development and statistical analysis; and the dissemination of results.

Protocols for sampling and analysis of determinands, many first developed in AL:PE are given in a MOLAR protocol manual (Wathne, 1996). The AL:PE chemical AQC programme has been maintained and developed to better address nitrogen and phosphorus determinations and extended to outside laboratories. Furthermore a system of sample exchange and workshops has been instigated to develop and extend the programme of taxonomic harmonization established in AL:PE.

A centralized database is being established and maintained by the Botanical Institute at the University of Bergen. This includes, in a consistent and structured way, all the available geographical, climatic, catchment environmental, lake chemical, lake physical, lake biological, and sediment chemical and biological data. The AL:PE/MOLAR database will be expanded and links with other international scientists encouraged, through presentation of MOLAR findings at national and international conferences and by promoting an international conference on remote mountain lake ecosystems under the auspices of the MOLAR Steering Group.

Further information may be found on the MOLAR Homepage on the Internet: <http://prfdec.natur.cuni.cz/hydrobiology/molar/welcome.html>.

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The influence of flow variation on the ability to evaluate the biological health of headwater streams

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Abstract Benthic invertebrate communities at 15 sites in a single drainage basin were examined over a 2-year period to relate streamflow to community structure and to evaluate the possibility of biomonitoring low-flow streams. Streams were evaluated for flow and habitat and were classed as intermittent, intermediate, or perennial. Each stream class had a characteristic community structure, although the differences among classes were more in relative abundances than in presence or absence. The intermittent stream community could be considered a subset of the perennial stream community. Indices of community structure were positively related to flow permanence, and the variability of indices increased as flow decreased. A biomonitoring system for low-flow streams would be less sensitive than for permanent streams but is possible if appropriate reference conditions for low-flow streams can be developed.

INTRODUCTION

First- and second-order streams comprise the greatest percentage of total stream length in a watershed and are significant contributors of nutrients, organic material, and inorganic sediments, as well as determinants of flow characteristics to downstream reaches. The health of these small water courses, variously described as ephemeral, intermittent, or temporary, is critical to the health of all waters in a watershed, and these headwaters are deserving of being included in monitoring efforts. Headwater streams have not been well studied; those that have show a wide variation in abiotic characteristics (Williams & Hynes, 1977) but a surprisingly similar fauna to that of more permanent streams (Delucchi, 1988; Boulton & Suter, 1986). The objective of this paper is to investigate the potential to use benthic invertebrates as biomonitors of headwater streams.

METHODS

The Brush Creek Watershed consists of 21 270 ha in southwestern Missouri, USA. The main stream of the watershed is Brush Creek, a small fourth-order tributary of the Sac River within the Osage River basin. Agriculture is the primary land use with 5% being row cropland, 48% pasture, 45% woodland, and 2% other. We studied 15 stream sites within the watershed which represented a variety of hydrological

conditions. Small streams in this region may become intermittent in late summer and autumn when riffles cease to flow and pools become isolated. The degree of intermittency varies from year to year depending upon the vagaries of precipitation and temperatures. The watershed appeared typical of most in this region where land-use effects on streams are evident yet the physical integrity of the channels and riparian vegetation are reasonably good. Channels are generally narrow, 1–4 m wide, with a chert gravel substrate.

Three classes of stream permanence were developed based upon late summer mean discharge and water depth in riffles. Discharge, water quality measurements, habitat measurements, and invertebrate collections were taken in September of 1995 and 1996, after extended uninterrupted low flow.

Macroinvertebrates were collected, identified, and recorded from riffles, pools, and rootmats according to established protocols (Rabeni *et al.*, 1997). Four indices or “metrics” of community structure were selected as measurements of stream health that change in response to environmental conditions (Plafkin *et al.*, 1989). The **EPT Index** is the sum of all Ephemeroptera, Trichoptera, and Plecoptera taxa; **Total Taxa** is number of taxa present; **Biotic Index** summarizes the overall pollution tolerance of the benthic arthropod community; and **Shannon Diversity Index** is a measure of community composition. Detrended Correspondence Analysis was used to examine the relative similarity of benthic invertebrate communities from all sites.

RESULTS

Grouping sites by the index of streamflow resulted in five permanent, four intermediate, and three (1995), or six (1996) intermittent sites. The Detrended Correspondence Analysis effectively separated the three stream community types for both years (Fig. 1). Because this ordination spatially relates actual species composition of each community to all other communities, and because there is very little interspersation of data points representing communities from different stream types, it is evident that each stream type has distinctive invertebrate community structure. It also appears that the permanent sites are most like each other, while the intermittent communities have the greatest intergroup differences, especially in 1996.

Community composition

We first examined the fauna to determine whether taxa were restricted to a particular stream type or whether they were widely distributed throughout the watershed (Table 1). Only a single taxa was found exclusively in any one stream category, although about 75% of the taxa showed definite distributional patterns. The insects *Tricorythodes* (Ephem.), *Enallagma* (Odon.), and *Dubiraphia* (Col.) were heavily situated in the permanent streams in both years. Taxa found in greater relative abundances in intermittent streams were often midges of the family Chironomidae (60% in 1995 and 33% in 1996). Several taxa “tended” toward more permanent situations—the insects *Chimarra* (Trich.), *Stenonema pulchellum* (Ephem.), *Tanytarsus* (Dipt.), and the amphipod *Hyaella*—while fewer tended toward

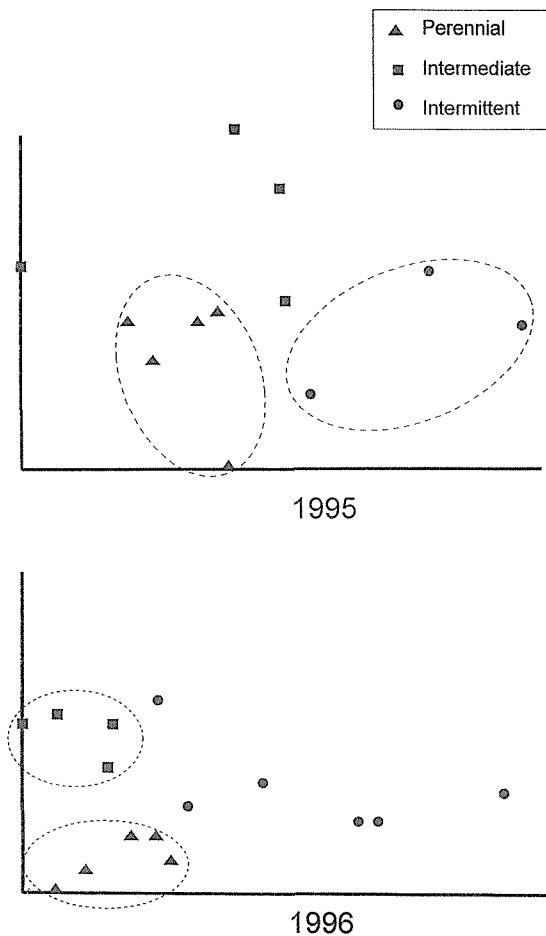


Fig. 1 Detrended correspondence analysis of each stream site from the Brush Creek basin.

intermittency—the Gastropoda and the water penny *Psephenus*. Most other taxa had widespread distributions without clear links to the flow regime. Thus, the community differences evident from the ordinations (Fig. 1) related more to relative abundances of the invertebrates than to presence and absence of taxa.

Biomonitoring metrics

Metric values were “best” for the permanent class streams followed by the intermediate streams and “worst” for the intermittent streams (Table 2). In the majority of cases the mean values for each stream type were significantly different from each other, with the greatest differences among stream types seen in total taxa and the EPT metric.

A common biomonitoring approach is the establishment of reference conditions developed from a group of reference streams, against which test streams are

Table 1 Percentage composition (>2% of total numbers) of invertebrate taxa collected from each stream class. X = present at one or more streams but comprise <2% of total; P = permanent, M = intermediate and I = intermittent.

	Stream class:		
	P	M	I
1995			
<i>Dubiraphia</i>	8.73	X	X
<i>Enallagma</i>	2.97	X	X
<i>Hydracarina</i>	2.10	X	
<i>Stenonema pulchellum</i>	2.90	X	
<i>Tricorythodes</i>	7.87	X	
<i>Chimarra</i>	3.14	12.45	
<i>Hyaella azteca</i>	10.29	2.22	X
<i>Polypedilum</i>	2.18	5.68	
<i>Tanytarsus</i>	2.90	4.22	X
<i>Cheumatopsyche</i>	X	2.24	
<i>Helicopsyche</i>	X	2.44	X
<i>Rheotanytarsus</i>	X	6.78	
<i>Thienemannimyia</i> gr.	X	3.62	X
<i>Psephenus</i>	X	4.95	2.45
<i>Stenacron</i>	X	4.86	8.97
<i>Ablabesmyia</i>	X		3.20
<i>Bactrurus</i>			2.05
<i>Dicrotendipes</i>	X	X	2.01
<i>Microtendipes</i>	X	X	3.45
<i>Oligochaeta</i>	4.16	X	8.28
<i>Caenis</i>	10.14	4.23	6.42
<i>Paratendipes</i>	4.34	10.32	6.72
<i>Stenelmis</i>	3.53	3.17	4.01
<i>Stenonema femoratum</i>	3.10	3.87	30.71
1996			
<i>Argia</i>	2.32	X	X
<i>Choroterpes</i>	2.22	X	X
<i>Dubiraphia</i>	8.76	X	X
<i>Enallagma</i>	2.08	X	X
<i>Hyaella azteca</i>	4.77	X	X
<i>Tanytarsus</i>	3.28	X	X
<i>Tricorythodes</i>	4.44	X	
<i>Polypedilum</i>	5.19	6.43	X
<i>Stenelmis</i>	6.62	3.85	X
<i>Stenonema pulchellum</i>	2.70	8.57	
<i>Baetis</i>	X	3.41	
<i>Cheumatopsyche</i>	X	2.52	
<i>Chimarra</i>	X	18.48	
<i>Psephenus</i>	X	6.86	
<i>Rheocricotopus</i>	X	3.69	
<i>Gastropoda</i>	X	3.05	4.26
<i>Allocrangonyx</i>		X	5.64
<i>Chironomus</i>	X	X	2.37
<i>Hydroporus</i>	X		8.26
<i>Orconectes</i>	X	X	2.68
<i>Paratendipes</i>	3.12	X	8.33
<i>Caenis</i>	13.46	5.39	14.44
<i>Oligochaeta</i>	6.04	2.78	26.26
<i>Stenacron</i>	2.30	5.33	2.75
<i>Stenonema femoratum</i>	2.85	2.69	6.27

evaluated. For the Brush Creek basin reference criteria developed from perennial streams would be inappropriate for non-perennial streams, but it should be possible to develop reference conditions for each stream category. A monitoring index or metric should possess low variation in reference situations so that deviations from reference conditions can be detected. If we consider our values to be the "reference condition" for each of the stream categories, we might now inquire how much the values must change to show a "significant" deterioration or improvement in stream quality? One way to calculate this is to use the metric's mean value and variation in a rearrangement of the *t*-test formula which is given in the caption of Table 2. Table 2 gives the approximate change necessary to be statistically different using a single sample comparison to a reference situation for the four metrics for each stream category. A trend is apparent for three of the four metrics: that is, a similar percent change may be detected in permanent and intermediate streams, but a much larger change is needed to detect a significant change in the intermittent streams. Because of the greater natural variation of metrics from intermittent streams, a large change in either degradation or improvement must occur before it is detected by the benthic invertebrates.

Table 2 Mean values and coefficient of variation for four biomonitoring metrics from the three stream classes and the percent change that would be necessary to be significantly different. Values are grand means for 1995 and 1996. P = permanent, M = intermediate, and I = intermittent; *N* = sample size, $\bar{X}$ = mean, *CV* = coefficient of variation. The final column was calculated using the formula $p^2 = (CV^2 k)/N$. Where *N* is the number of samples (1 in this case), *k* is a constant that varies with alpha and statistical power, *CV* is the coefficient of variation (sd/mean) of metric values, and *p* is the change expressed as a proportion of the mean (Parkinson *et al.*, 1988). We used alpha = 0.05 and 80% statistical power which gave a *k* = 12.57.

Metric	Stream category	<i>N</i>	$\bar{X}$	<i>CV</i>	% change needed to be detected
Total taxa	P	10	71	12.4	18
	M	8	58	12.8	18
	I	9	22	32.1	28
EPT	P	10	18	21.5	23
	M	8	14	20.5	23
	I	9	4	63.1	40
Biotic Index	P	10	6.2	5.3	11
	M	8	5.4	8.2	14
	I	9	7.0	11.6	17
Shannon Diversity	P	10	3.3	5.0	11
	M	8	3.0	4.8	5
	I	9	2.0	22.3	24

DISCUSSION

Biomonitoring should have a sound ecological basis so as to have generality in various situations. Invertebrates in more "temporary" habitats may have adaptations that include a drought resistant life history stage, the ability to migrate from drying conditions, or an alteration of the life cycle allowing fast growth and early emergence (Williams & Hynes, 1977). However, most studies that have examined

this question have shown intermittent streams not to have a unique fauna (Feminella, 1996) and few species appear specifically adapted to temporary streams although distinct seasonal cycles related to flow were noted (Boulton & Lake, 1992). In this study there were relatively few taxa not collected either year in the intermittent streams but collected from more permanent stream types. Some could be considered as requiring moving water, such as the net spinning caddisflies *Chimarra* or *Cheumatopsyche*, or other "typical" riffle-dwelling taxa such as *Baetis*. For others, however, such as the mayfly *Tricorythodes*, reasons for their absence from intermittent streams were not as obvious. Overall, differences in the faunas among streams of differing permanence are more commonly due to densities or relative abundances of taxa. Few biomonitoring metrics use actual densities so this fact should not be important. It would be important not to have an intermittent stream fauna composed mostly of taxa insensitive to anthropogenic assaults. This may be partially the case because the biotic index, based on tolerances of taxa is substantially greater for intermittent streams than for the more permanent stream types (Table 2). The mean tolerance value for invertebrates concentrated in permanent streams was 5.6, while for taxa concentrated in intermittent streams it was 6.4.

This study shows that invertebrates inhabiting intermittent and low-flow streams have the potential to be effectively monitored, either to investigate trends over time or for a comparison to a reference condition. However, some cautions are advised. The designations of flow type are subject to interpretation and can change over time, either due to year-to-year variations in precipitation or more long-term trends. Temporal annual sequences in fauna are important in temporary streams (Feminella, 1996), and antecedent hydrologic conditions affect species composition (Boulton & Lake, 1992). The variability of indices in intermittent streams is greater than for permanent streams, and a greater change in the benthic fauna would have to occur to be noted. However, the importance of these streams to the overall health of the system makes the development of a monitoring system an important endeavour.

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The Val Roseg project: temporal and spatial patterns of benthic algae in an Alpine stream ecosystem influenced by glacier runoff

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Abstract We studied the spatial and temporal distribution of benthic algae in an Alpine stream ecosystem strongly influenced by glacier runoff. Algal biomass, as chlorophyll *a*, was measured at several stations in the main channel and in groundwater-fed channels in a glacial flood plain. Mean algal biomass increased downstream from the glacier (from 0.3 to 9.3 mg chl. *a* m⁻²) during summer high flow. The inverse trend was observed in autumn (from 138 to 12 mg chl. *a* m⁻²). The chrysophyte *Hydrurus foetidus* dominated the phytobenthic community close to the glacier. Algal biomass in groundwater channels exhibited high spatial variability. For example, in August mean values in different groundwater-fed channels ranged from 4 to 218 mg chl. *a* m⁻². During summer high flow or after major spates, the groundwater-fed channels are presumably the only sites of significant autochthonous energy fixation in this glacial stream network.

INTRODUCTION

Biologically mediated energy flow through lotic ecosystems is fueled by primary production in the wetted channel and allochthonous inputs of organic matter produced on the flood plains and uplands. The relative contributions of these resources to the energy flow of lotic ecosystems vary significantly along the river continuum (Vannote *et al.*, 1980). High Alpine headwaters drain catchments where terrestrial vegetation is scarce or even lacking. Therefore, input of allochthonous organic matter is expected to be small (Millner & Petts, 1994; Ward, 1994). In these systems, photosynthesis of benthic algae is thought to be a major energy source supporting the heterotrophic community (Lavandier & Décamps, 1984). However, information on energy flow through an Alpine stream ecosystem is scarce.

Glacial streams in the Alps are presumably very harsh environments for benthic algae. High turbidity, snow cover, abrasion by suspended solids and bed-load transport during summer high flow (ice melt season) are thought to limit the formation of algal biofilms. In this paper, we present the first results of an investigation focused on temporal and spatial distribution of benthic algae in a glacier stream of the Swiss Alps. The investigation is part of a comprehensive ecological investigation of a high Alpine stream ecosystem (Tockner *et al.*, 1997; Ward *et al.*, 1997).

HEADWATER SYSTEM

The study system is the Roseg River in the eastern part of the Swiss Alps (Fig. 1). About 30% of the drainage basin (area 66.5 km²) is covered by glaciers. Meltwater

from the Tschierva and Roseg glaciers is the primary water source of the Roseg River. Five major reaches characterize the study system, namely a partly braided reach in the recent proglacial area of the Tschierva Glacier (length 500 m), a lake outlet downstream from the Roseg glacial lake (length 900 m), a single-thread channel incised in glacial till (length 900 m), the main glacial flood plain (length 2800 m) with various channels of different hydrology and chemistry (Tockner *et al.*, 1997), and a reach that is mainly constrained by steep valley slopes (length 7200 m). See Tockner *et al.* (1997) for details on hydrology, physics and chemistry.

METHODS

Discharge (Q) data were obtained from a gauging station of the Swiss Hydrological and Geological Survey close to the end of the drainage basin (river km 0.85). Water

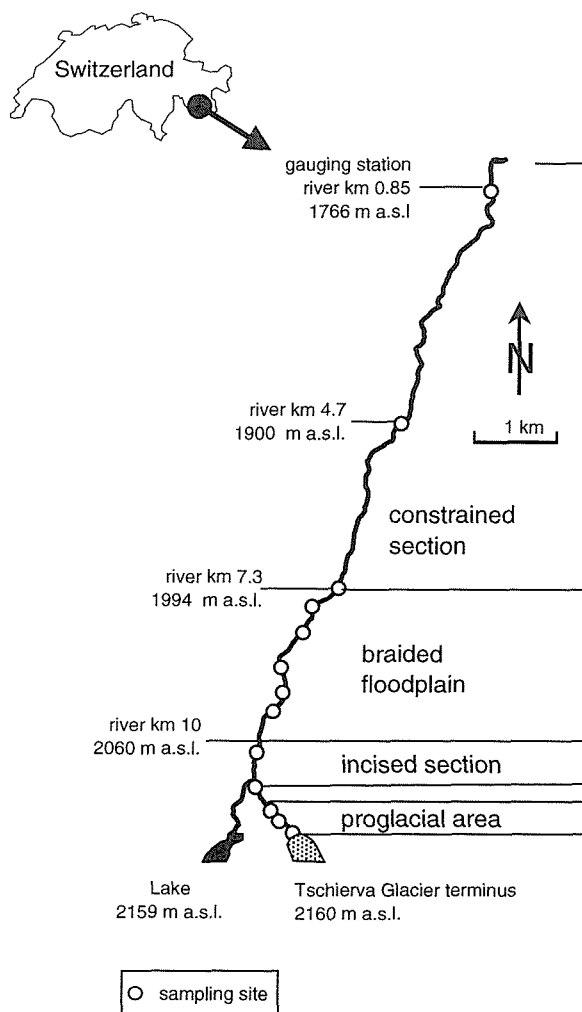


Fig. 1 Location and description of the Val Roseg.

temperature was continuously monitored at selected locations with temperature loggers. Turbidity was measured with a portable turbidity meter. Subsurface photosynthetically active radiation (PAR) was measured using a quantum sensor in 0.1-m depth intervals and vertical PAR attenuation coefficients were calculated by linear regression of $\ln(\text{PAR}(z))$ against depth (z).

Benthic algae were sampled at several stations along the main channel and in different channel types of the flood plain. From August 1997 to February 1997 five stones were collected at random at each site (Leopold, 1970). After February the number of stones collected was doubled. The stones with attached algae were frozen and transported to the laboratory in cooling boxes. Samples were processed as described by Uehlinger *et al.* (1996) and chlorophyll *a* was determined according to Meyns *et al.* (1994).

RESULTS AND DISCUSSION

The physical habitat template

In July and August 1996 discharge varied between 3.5 and 23.5 m³ s⁻¹, and subsequently declined to *c.* 0.1 m³ s⁻¹ by January 1997 (Fig. 3(a)). Discharge began to rise again in May 1997. In the proglacial reach, bed-load transport could be detected during summer high flow from the sound of rolling stones. From December 1996 to mid April 1997 the upper part of the main channel in the flood plain section (km 8.7–10) was dry. Flow continued in the incised reach, which was fed by the lake outlet, and presumably in the proglacial section (Ward *et al.*, 1997).

During summer, turbidity in the main channel varied between 115 and 1500 NTU (Table 1), corresponding to an attenuation of PAR by 80% and >99.99% in a water column of 0.5 m depth. Turbidity decreased as flow declined. In September 1996 turbidity was actually lower close to Tschierwa Glacier than in the incised reach or in the flood plain. During winter, upper parts of the main channel (upstream of river km 7.2) were under a snow cover of >1 m thickness (PAR measurements showed that a snow pack of 0.5 m attenuated 99.5% of incident PAR). Snow covered *c.* 95% of the main channel below river km 8.7 between December and March 1997. Mean daily temperatures in the main channel varied between 0°C and 1°C below the glacier, 0°C and 4.2°C in the middle of the flood plain, and 0.2°C and 7.4°C at the end of the drainage basin (Fig. 2).

Compared to the main channel, perennial groundwater-fed channels in the flood plain provide a rather benign environment for benthic algae. The more constant discharge lacked the capacity to move bed sediments. Turbidity was <10 NTU. Therefore, PAR attenuation was small. Some of these channels did not develop ice cover. Temperatures were higher than in the main channel of the flood plain (Fig. 2).

Table 1 Average turbidity (NTU) in three reaches of the main channel measured on several sampling occasions. n.d. = no data.

Reach	6/8/96	23/8/96	29/9/96	3/11/96	14/5/97	10/7/97
Flood plain	82	1005	36	62	9	197
Incised reach	124	928	57	64	22	300
Proglacial area	177	n.d.	13	7	17	261

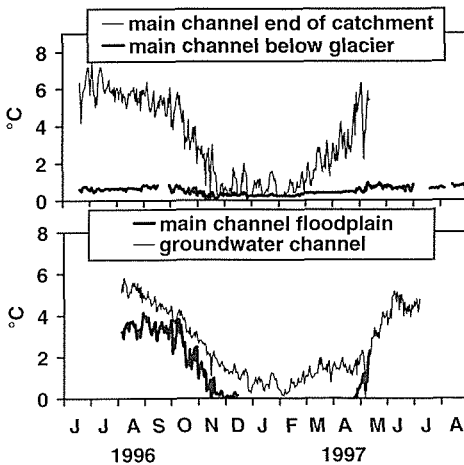


Fig. 2 Mean daily temperature at three stations in the main channel and in a groundwater channel of the flood plain.

Temporal and spatial variability of benthic algal biomass

In the main channel algal biomass reflected the seasonally changing environmental conditions. In August 1996, during summer high flow ($6\text{--}13\text{ m}^3\text{ s}^{-1}$, Fig. 3(a)), biomass averaged $0.3\text{ mg chl. } a\text{ m}^{-2}$ in the unstable channels below the glacier and increased to $9.3\text{ chl. } a\text{ m}^{-2}$ in the flood plain (Fig. 3(b),(c)). This trend was reversed at the end of September when biomass decreased downstream from 138 to $11.3\text{ mg chl. } a\text{ m}^{-2}$. In the flood plain and the constrained section below the flood plain, biomass peaked in November with 71 and $87\text{ mg chl. } a\text{ m}^{-2}$, surprisingly high values for unenriched streams (Biggs, 1996). By March 1997 biomass had decreased to 18 and $10\text{ mg chl. } a\text{ m}^{-2}$. In May, when the main channel was open, turbidity was low, and flow relatively moderate ($c. 2\text{ m}^3\text{ s}^{-1}$), biomass ranged only from 4 to $6\text{ chl. } a\text{ m}^{-2}$ between the glacier and the end of the drainage basin. The spate of 29 June 1997 ($Q_{\max} = 34\text{ m}^3\text{ s}^{-1}$) scoured most algae from the main channel; 10 days after this event biomass values ranged only between 0 and $1.1\text{ mg chl. } a\text{ m}^{-2}$. The chrysophyte *Hydrurus foetidus* was the predominant algal species near the Tschierwa Glacier (Fig. 4(a)). In the flood plain, the relative abundance of *H. foetidus* was still 18% to 43%, and diatoms (mainly *Fragilaria arcus*, *Achnanthes sp.* and *Cymbella sp.*) and cyanobacteria (mainly *Homeothrix sp.* and *Chamaesiphon sp.*) contributed 6% to 20% and less than 1% to 17%, respectively (Fig. 4(b)).

In groundwater-fed channels, the biomass of benthic algae reached maximum values in late autumn and winter (Fig. 3(e)). In March 1997 biomass was low with values similar to those in the perennial section of main channel. This presumably reflects the fact that habitat conditions (chemistry, turbidity, temperature and flow) in the main channel and groundwater-fed channels were quite similar during this period (Tockner et al., 1997). The algal community of groundwater channels was dominated by diatoms and cyanobacteria (Fig. 4(c)), with the relative abundance of *H. foetidus* ranging from 6% to 20%. Taxa richness in groundwater channels was higher than in the main channel (Bürgi, unpublished data).

The spatial variability of benthic algae at the microhabitat level (single stone) was often high. Coefficients of variation ranged from 20% to 316%. During summer

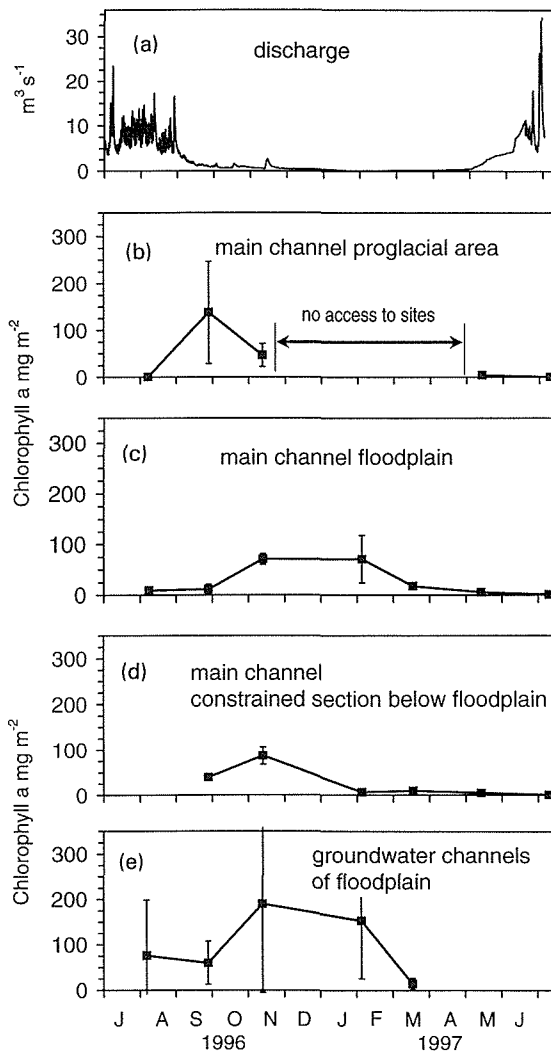


Fig. 3 (a) Discharge in the main channel, (b)–(d) biomass of benthic algae in the main channels and (e) in groundwater channels of the flood plain.

high flow differences between sites along the main channel seemed to be relatively small (e.g. August 1996 and July 1997) but became quite large in autumn (Fig. 3). The variability between different groundwater channels was high except in March 1997 (Fig. 3(e)).

CONCLUSION

In glacial streams of the Alps, the availability of light and flow are presumably the most important determinants of algal development. Biomass of benthic algae in the main channel of the Roseg River glacial stream ecosystem was low during summer high flow, apparently due to sediment transport, shear stress and high turbidity. In

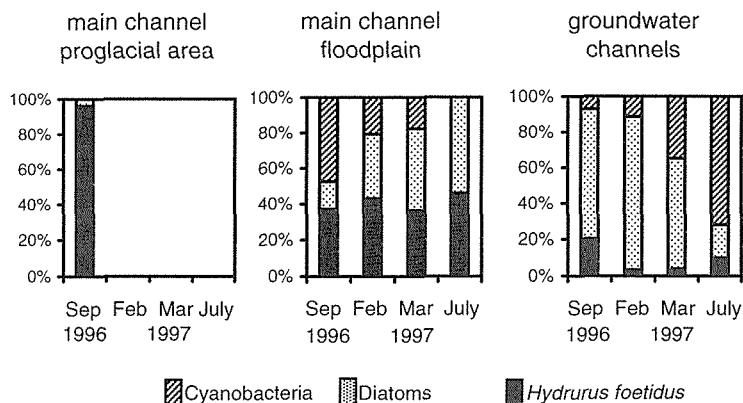


Fig. 4 Relative abundances of phyto-benthic taxa in the main channel (proglacial area and flood plain) and in groundwater channels of the flood plain.

the main channel there were apparently periods of time (late spring and early autumn), during which flow and light could lead to the accrual of benthic algae. However, biomass remained at low levels during spring in the main channel. Algal biomass was relatively high in flood-plain channels that are not connected to the main channel. These channels are presumably the only sites of significant autochthonous energy inputs into the ecosystem during summer high flow or after major spates.

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The Val Roseg project: habitat heterogeneity and connectivity gradients in a glacial flood-plain system

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Abstract The Val Roseg Project is a holistic ecological study of an alluvial flood plain in the Swiss Alps. Major geomorphic features consist of a proglacial reach below the terminus of Tschierva Glacier, the outlet stream of a glacial lake at the terminus of Roseg Glacier, and the glacial flood plain. Here we describe the habitat heterogeneity of the glacial flood plain, which is 2.6 km long and up to 510 m wide. Transects were established every 150–200 m to investigate gradients of turbidity, temperature, nutrients, organic matter, and surface water–groundwater interactions. Six distinct channel types have been identified, based on the correspondence between connectivity and physico-chemical characteristics. Seasonal and diel variations in discharge are marked and play a prominent role in structuring the habitat template. The entire flood plain shifts from dominance by glacial meltwater in summer to a groundwater-controlled system in winter.

INTRODUCTION

Alpine stream ecosystems have been proposed as sensitive indicators of climate change and anthropogenic impacts (McGregor *et al.*, 1995), yet relatively little is known of their ecology (Ward, 1994). Despite interest in high mountain streams by European limnologists early in this century (Steinmann, 1907; Thienemann, 1912; Steinböck, 1934; Léger, 1937), we are not aware of a single holistic ecological study that has been conducted on a glacial flood plain. Therefore, we initiated a comprehensive ecological investigation of a pristine glacial flood-plain system in the Swiss Alps (Fig. 1) to determine how spatio-temporal heterogeneity (a) contributes to biodiversity patterns of lotic zoobenthos, groundwater fauna and periphyton in a harsh environment and (b) influences ecosystem processes such as productivity and decomposition dynamics. In this paper we focus on habitat heterogeneity and connectivity for the surface waters of the flood-plain complex.

DESCRIPTION OF STUDY AREA

Val Roseg is situated in the Bernina Massif of the Swiss Alps (Fig. 2). In the upper Roseg catchment (49.5 km², 42% covered by glaciers) elevations range from 1990 m a.s.l. at the downstream end of the glacial flood plain to 4049 m at the top of Piz

The Val Roseg Project

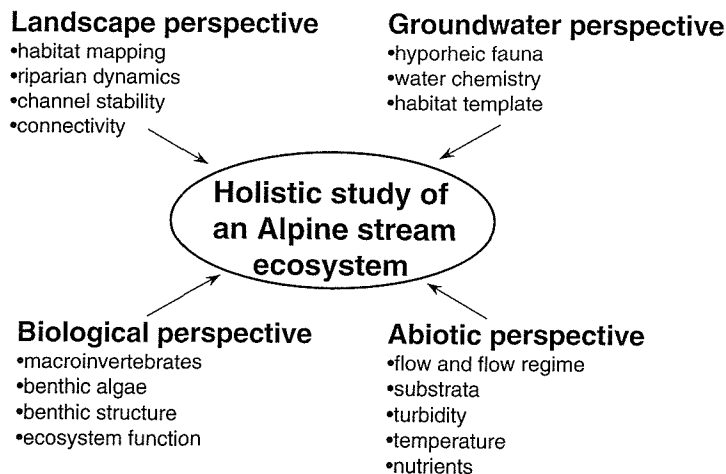


Fig. 1 The multifaceted nature of the Val Roseg Project.

Bernina. The Roseg River is a second-order tributary of the River Inn, a major tributary of the Danube. The primary water source is meltwater from two valley glaciers: the Tschier Glacier (6.2 km²) and the Roseg Glacier (8.5 km²). Five major stream reaches are apparent (Fig. 2). This paper focuses on the lotic channel types of the flood plain, which is 2.6 km long and up to 510 m wide and has a mean slope of 3.3%. The width of the main channel in the flood-plain reach ranged between 10 and 26 m during early summer flow conditions (*c.* 5 m³ s⁻¹).

MAPPING, SAMPLING AND ANALYSIS

In summer 1996 the positions of individual channels, springs, and marshes were determined along 17 transects spaced 150–200 m apart along the length of the flood plain. The width, depth, and wetted perimeter of each channel were recorded along each transect. Aerial photographs (infrared and colour images, spatial resolution 0.1 m) were taken on 3 September 1996 from a helicopter at 500 m above ground level.

Water temperatures were recorded continuously at selected locations with temperature loggers. Other physico-chemical samples were collected using standard procedures and measurements were conducted employing analytical methods described in Tockner *et al.* (1997).

RESULTS AND DISCUSSION

Only selected parameters, from a suite of physico-chemical variables are referred to in this paper. To portray the habitat heterogeneity of the glacial flood plain, we present a typology of channel (habitat) types and examine aspects of flow dynamics, the temperature regime, and spatio-temporal patterns of turbidity and specific conductance.

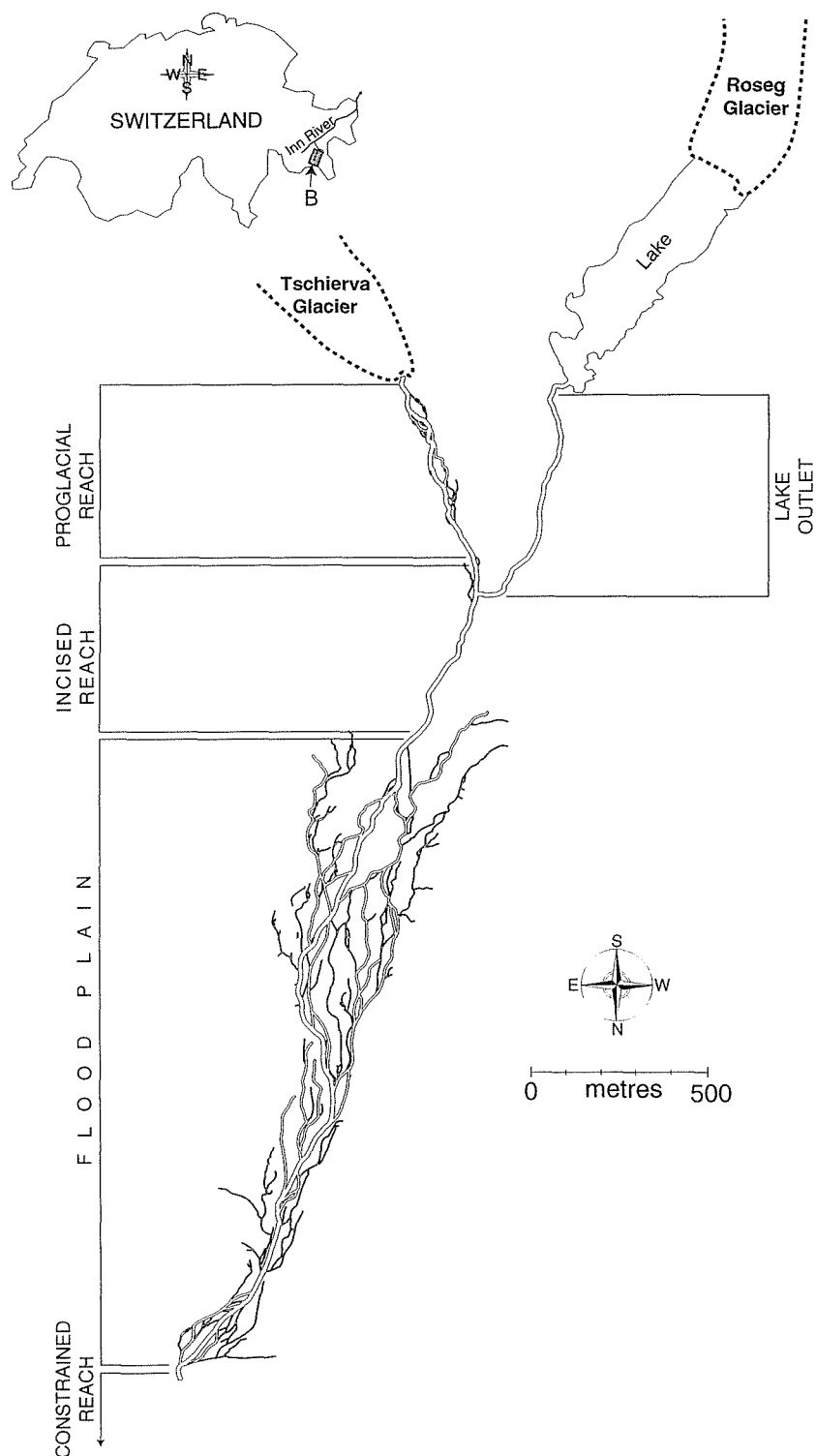


Fig. 2 Location of the upper Val Roseg catchment, showing the five major stream reaches (modified after Tockner *et al.*, 1997).

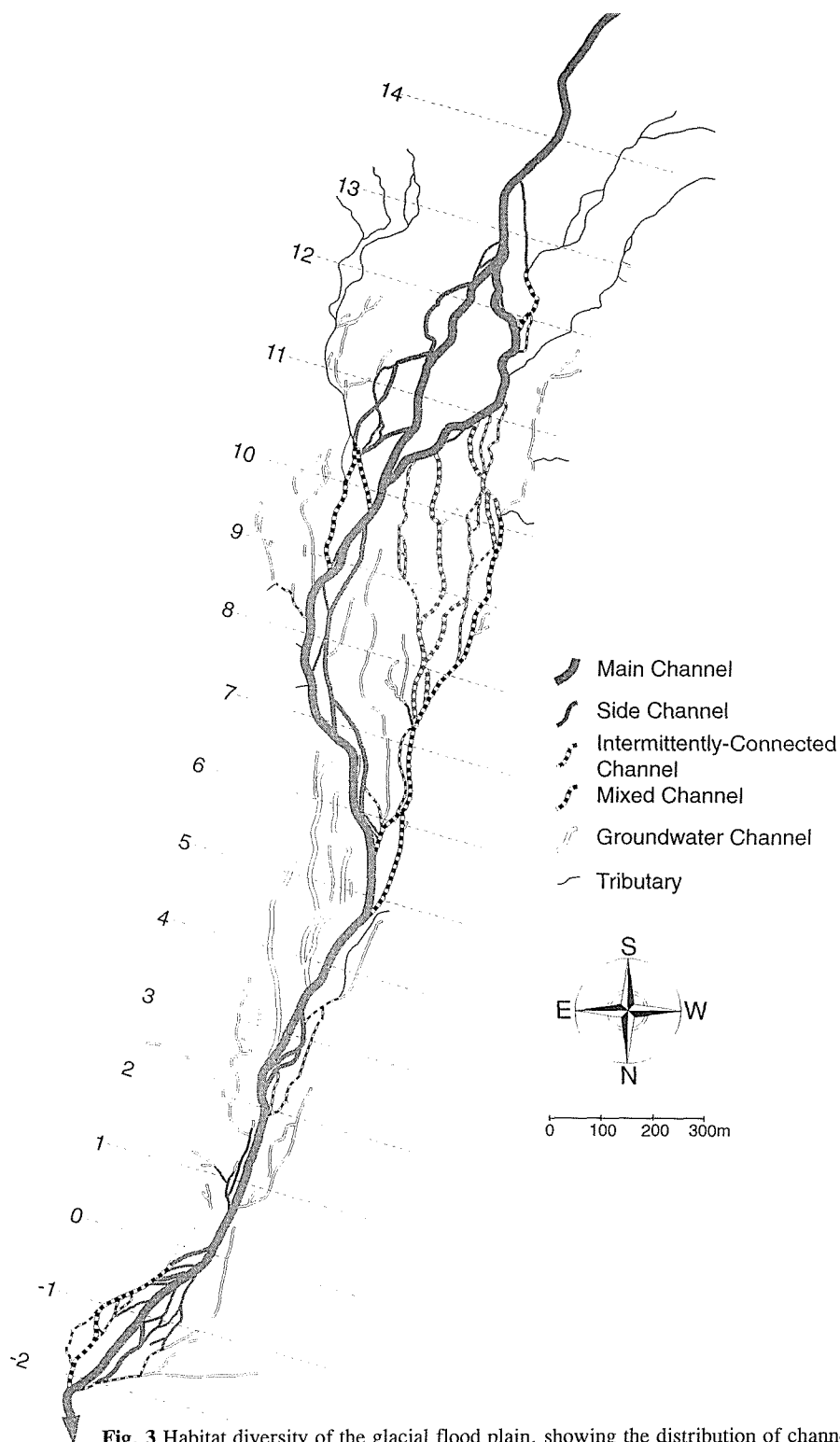


Fig. 3 Habitat diversity of the glacial flood plain, showing the distribution of channel types (modified after Tockner *et al.*, 1997).

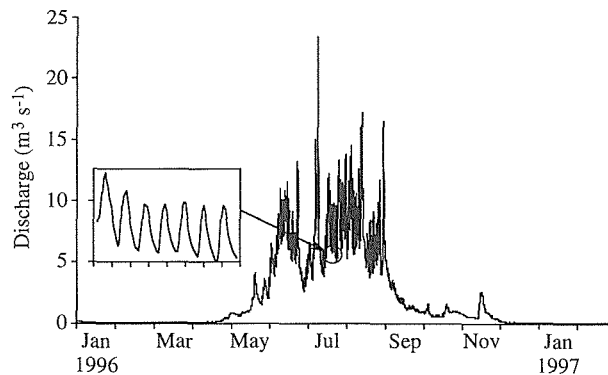


Fig. 4 The hydrograph of the Roseg River, showing seasonal and diel discharge patterns.

Channel typology

The glacial flood plain exhibited a much greater degree of habitat heterogeneity than expected, which was reflected by a diversity of channel types (Fig. 3). Based on the correspondence between connectivity and physico-chemical attributes, six major channel types were identified, as follows:

- (a) *Main channel (M)*—the primary channel (thalweg) draining the flood plain. Fed primarily by glacial meltwater.
- (b) *Side channels (S)*—eurhithric side channels connected to the main channel at their upstream and downstream ends; may not contain surface water under low flow conditions.
- (c) *Intermittently-connected channels (I)*—pararhithric side channels that are permanently connected to the main channel downstream and intermittently connected to the main channel upstream.
- (d) *Groundwater channels (G)*—fed by water upwelling from contiguous aquifers at their upstream end and connected to the main channel downstream.
- (e) *Tributaries (T)*—enter the flood plain as a surface water channel originating in the uplands; water sources are from hanging glaciers or hillslope aquifers.
- (f) *Mixed channels (X)*—contain a mixture of water derived from two or more of the other channel types.

Even these major channel types do not fully express the habitat heterogeneity of the glacial flood plain. Based on the specific water source (e.g. deep *vs* shallow groundwater; alluvial *vs* hillslope aquifer; subglacial groundwater), groundwater channels alone encompass several distinct subtypes (Ward *et al.*, 1998).

Flow dynamics

Seasonal and diel variations in discharge are marked (Fig. 4), as expected in a system fed by glacial meltwater, and play a prominent role in structuring the habitat template of the flood plain. Surface waters underwent dramatic cycles of habitat expansion, contraction, fragmentation, and reconnection on a seasonal basis (Fig. 5), an attribute of stream ecosystems that has received little attention (Stanley *et al.*, 1997). The

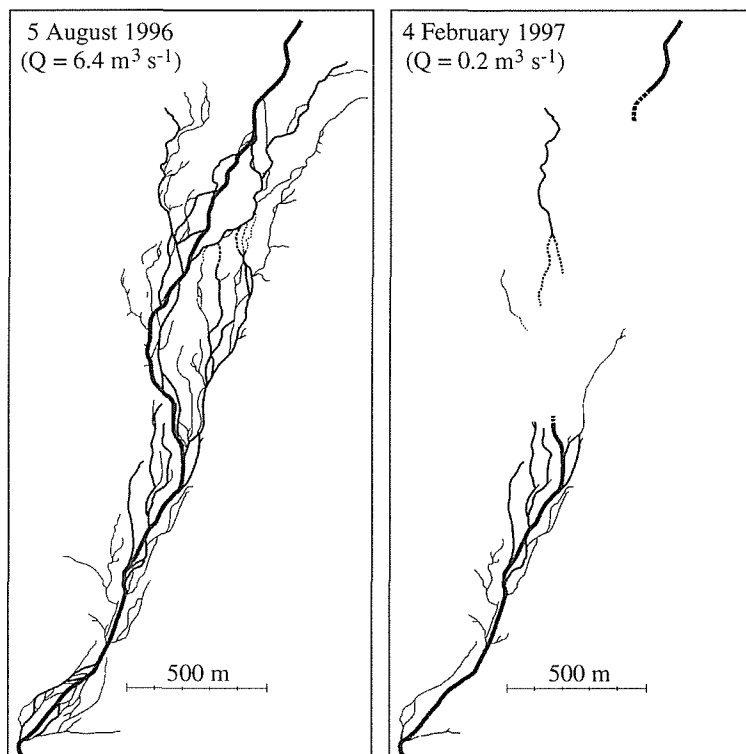


Fig. 5 The channel network of the flood plain during high summer flow (left) and low winter flow (right) (modified after Ward *et al.*, 1998).

number of channels per transect varied from 1 to 11 during high discharge in summer ($6.2\text{--}10.7\text{ m}^3\text{ s}^{-1}$), from 1 to 7 during low discharge in autumn ($0.9\text{ m}^3\text{ s}^{-1}$), and from 1 to 6 during the low discharge in winter ($0.2\text{ m}^3\text{ s}^{-1}$). Total channel length declined from 17.6 km during summer, to 14.2 km in autumn, to only 5.4 km in winter. The entire flood plain shifts from dominance by glacial meltwater in summer to a groundwater-controlled system in winter.

Temperature regime

A much greater degree of thermal heterogeneity was recorded than anticipated for a glacial flood plain (Table 1). The main channel, influenced most directly by glacial meltwater, had the lowest mean temperature (3.5°C) and the lowest diel amplitude

Table 1 Mean water temperatures ($^\circ\text{C}$) and diel amplitudes in four channel types, based on continuous readings from 5 August to 27 September 1996.

Channel type	Daily mean (range)	Diel amplitude (range)
Main	3.5 (2.8–4.1)	2.6 (0.5–5.7)
Intermittently-connected	5.4 (3.9–7.9)	4.6 (1.4–9.0)
Mixed	5.9 (5.0–7.2)	4.2 (0.9–7.2)
Groundwater	4.9 (4.1–5.8)	3.1 (0.9–5.3)

(2.6°C). The groundwater channel combined a relatively high mean temperature (4.9°C) with a low diel amplitude (3.1°C).

Turbidity and specific conductance

Turbidity and specific conductance were used as indicators of hydrological connectivity and water source (Fig. 6). These two parameters were inversely related along both

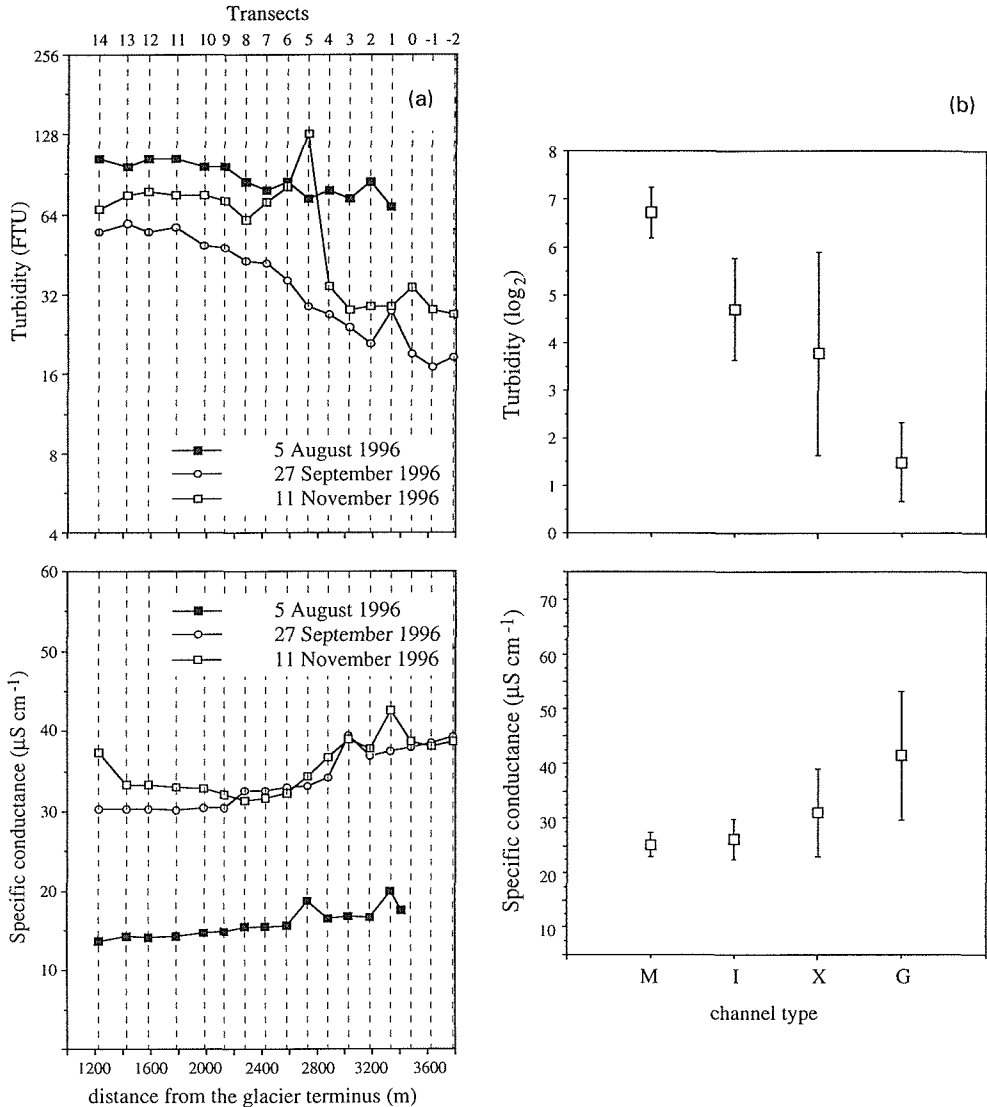


Fig. 6 (a) Longitudinal changes in turbidity and specific conductance in the main channel under high (August), medium (September) and low (November) flow conditions (modified after Ward *et al.*, 1998). (b) Turbidity and specific conductance as a function of channel type. Means ($\pm$ s.d.) of the average values for individual channels within each channel type. M = main channel; I = intermittently-connected channels; X = mixed channels; G = groundwater channels (modified after Tockner *et al.*, 1997).

temporal and spatial dimensions. For example, as discharge declined from summer to winter, turbidity decreased but specific conductance increased. The spatial patterns of these parameters also were inversely related, with turbidity decreasing and specific conductance increasing from the main channel type toward the groundwater channel type. Because groundwater channels do not have upstream surface connections with the main channel, they do not receive turbid glacial meltwater. In addition, groundwater-fed channels receive “older” water that has had more time to increase in ionic strength, whereas glacial meltwater typically has specific conductance values lower than $15 \mu\text{S cm}^{-1}$.

CONCLUSIONS

The surface waters of the Val Roseg flood plain exhibit a high degree of habitat heterogeneity, much of it attributable to the diversity of channel types. Preliminary analyses reveal an aquatic fauna that is also much more diverse and abundant than one would expect in a glacial environment that undergoes cyclic expansion and contraction. Groundwater channels, individually stable, yet collectively encompassing greater spatial heterogeneity than other channel types (Ward *et al.*, 1998), may play a major role in sustaining the ecological integrity of the flood-plain complex. Future research will investigate the ecological strategies that enable biotic assemblages to sustain populations under the harsh conditions experienced in glacial flood-plain habitats.

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